

From Linear Models to Neural Networks

Linear & logistic regression, neurons, MLPs, universal approximation

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ES 667 · Deep Learning · IIT Gandhinagar

Lecture 1 chose the loss; Lecture 2 expands the prediction rule

Keep from Lecture 1

Regression output $\rightarrow$ Gaussian model
 $\rightarrow$ MSE

Class probabilities $\rightarrow$ Bernoulli /
categorical model $\rightarrow$ cross-entropy

Change in Lecture 2

Start with one weighted sum, expose what it cannot represent, then let hidden layers learn nonlinear features.

$x \rightarrow f_{\theta}(x) \rightarrow$ output parameters $\rightarrow$ same Lecture 1 loss

We will not re-derive the losses; today is about making f_{θ} more expressive.

Outline

1. **Linear prediction** — weighted sums, batches, and output shapes
2. **One neuron** — add an activation to create a nonlinear feature
3. **The limitation** — see exactly why a linear model cannot solve XOR
4. **An MLP** — compose learned features across width and depth
5. **Practice** — trace one forward pass, count parameters, and make one update

One notation covers every prediction rule

Write $f_{\theta}(x)$ for the model's prediction at input x .

x : the input features

θ : all trainable weights and biases

f_{θ} : the rule that maps input to a score or prediction

Starting point — one weighted sum plus a bias:

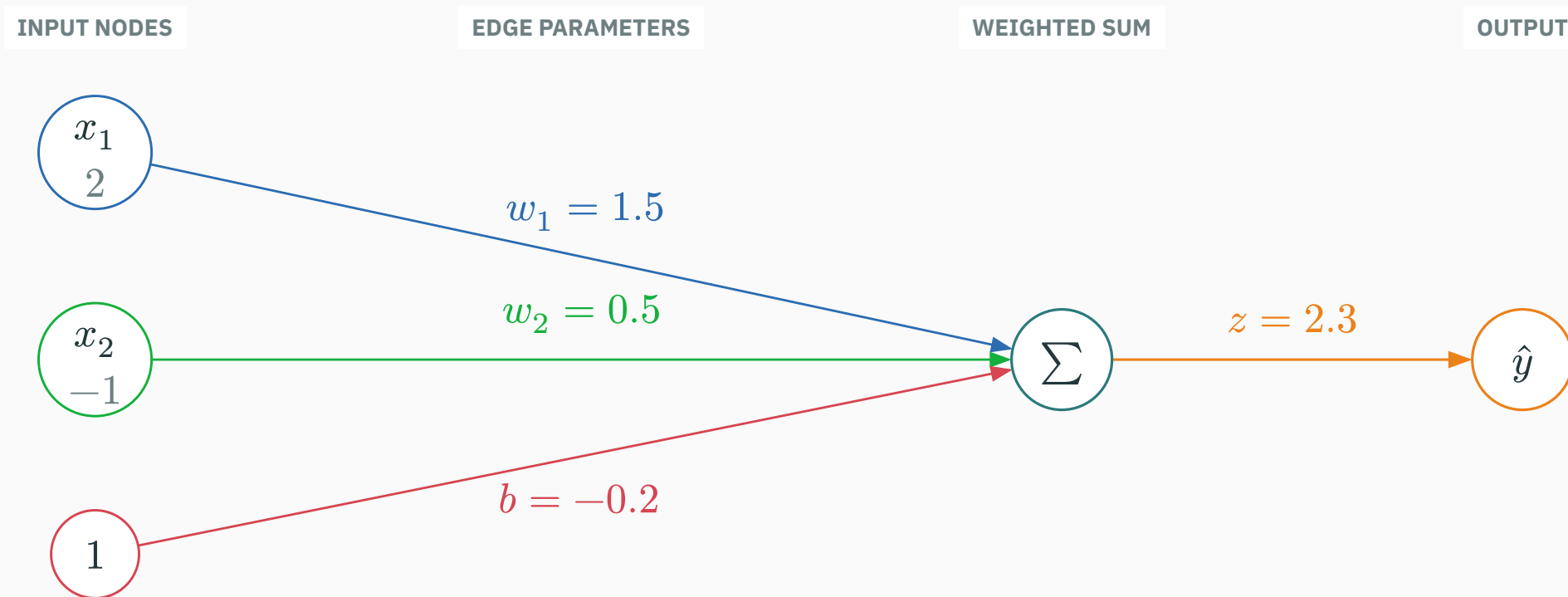
$$f_{\theta}(x) = w^{\top} x + b$$

We will build the network gradually; the full nested formula comes only after each part is concrete.

The linear starting point

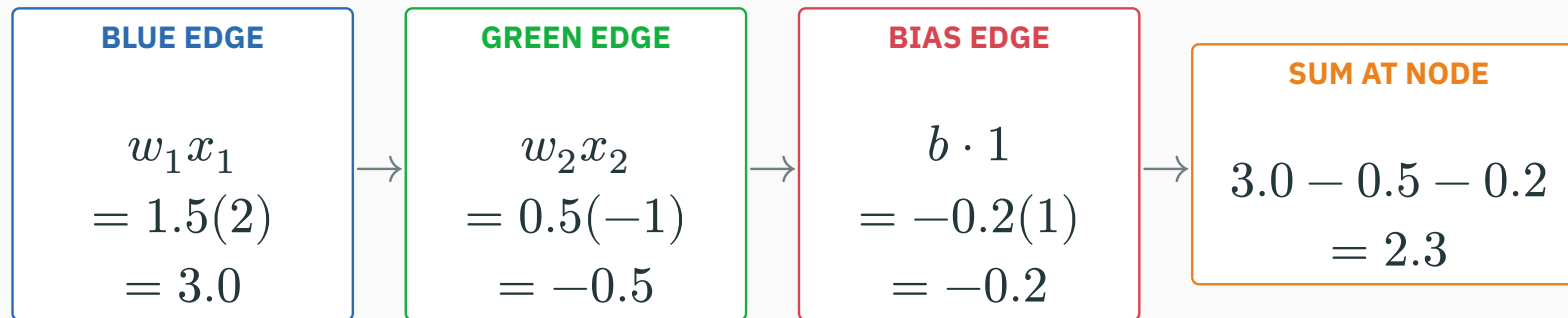
Start with one weighted sum

For one output, every input node sends a weighted value into one summation node.



The constant node 1 turns the bias into one more incoming edge: $b \cdot 1$.

Calculate every contribution entering the node



$$\hat{y} = \underbrace{1.5(2)}_{3.0} + \underbrace{0.5(-1)}_{-0.5} + \underbrace{(-0.2)(1)}_{-0.2} = 2.3.$$

Each edge contributes “weight × incoming value”; the node adds all contributions.

A batch is a labelled data matrix

example	feature x_1	feature x_2	target y
1	2	-1	2.4
2	0	3	1.0
3	-1	1	-1.1

DATA MATRIX X

$$X = \begin{pmatrix} 2 & -1 \\ 0 & 3 \\ -1 & 1 \end{pmatrix}$$

3 rows = examples
2 columns = features

TARGET VECTOR y

$$y = \begin{pmatrix} 2.4 \\ 1.0 \\ -1.1 \end{pmatrix}$$

one observed target
per example

Row i contains the input $x^{(i)}$ and its observed answer $y^{(i)}$.

One matrix multiplication predicts all three rows

Reuse $w = \begin{pmatrix} 1.5 \\ 0.5 \end{pmatrix}$ and $b = -0.2$ for every example.

$$\begin{aligned} Xw + b\mathbf{1} &= \begin{pmatrix} 2 & -1 \\ 0 & 3 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1.5 \\ 0.5 \end{pmatrix} + (-0.2) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 2.5 \\ 1.5 \\ -1.0 \end{pmatrix} + \begin{pmatrix} -0.2 \\ -0.2 \\ -0.2 \end{pmatrix} = \begin{pmatrix} 2.3 \\ 1.3 \\ -1.2 \end{pmatrix} = \hat{y}. \end{aligned}$$

ROW 1

$$1.5(2) + 0.5(-1) - 0.2 = 2.3$$

ROW 2

$$1.5(0) + 0.5(3) - 0.2 = 1.3$$

ROW 3

$$1.5(-1) + 0.5(1) - 0.2 = -1.2$$

$\hat{y}$ contains predictions; y contains observed targets. They are not the same object.

AUGMENT THE DATA MATRIX

$$\tilde{X} = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 3 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

 $\tilde{X}$

original feature columns
+ one constant column

STACK THE PARAMETERS

$$\theta = \begin{pmatrix} w_1 \\ w_2 \\ b \end{pmatrix} = \begin{pmatrix} 1.5 \\ 0.5 \\ -0.2 \end{pmatrix}$$

 θ

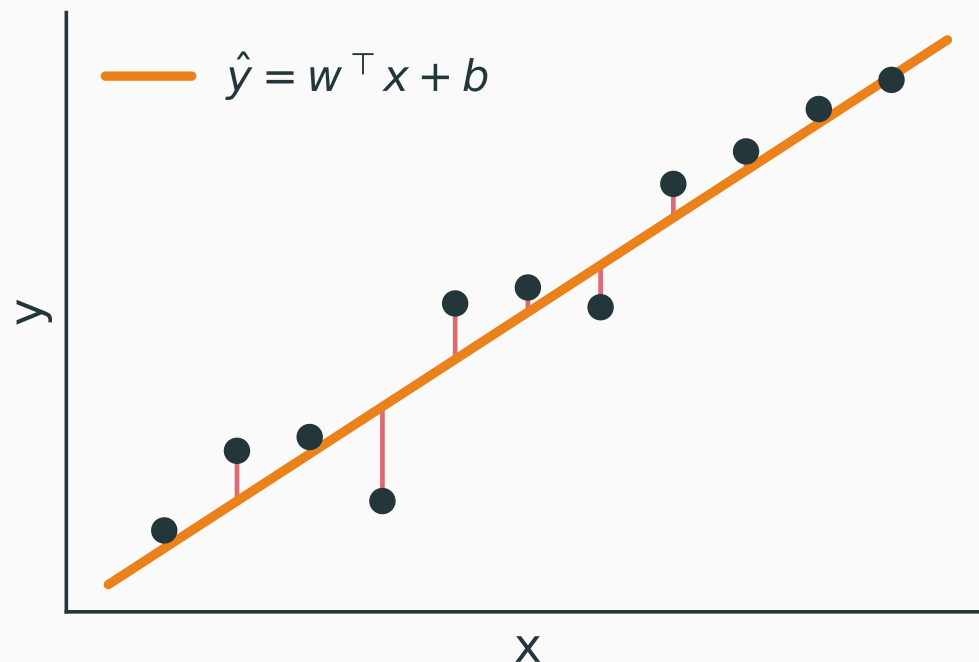
all weights
+ the bias

$$\hat{y} = \tilde{X}\theta, \quad (n \times (d + 1))((d + 1) \times 1) \rightarrow (n \times 1).$$

The column of ones is the batch version of the constant bias node on the previous network diagram.

For regression, one weighted sum gives one flat trend

$$\hat{y} = w^{\top} x + b$$

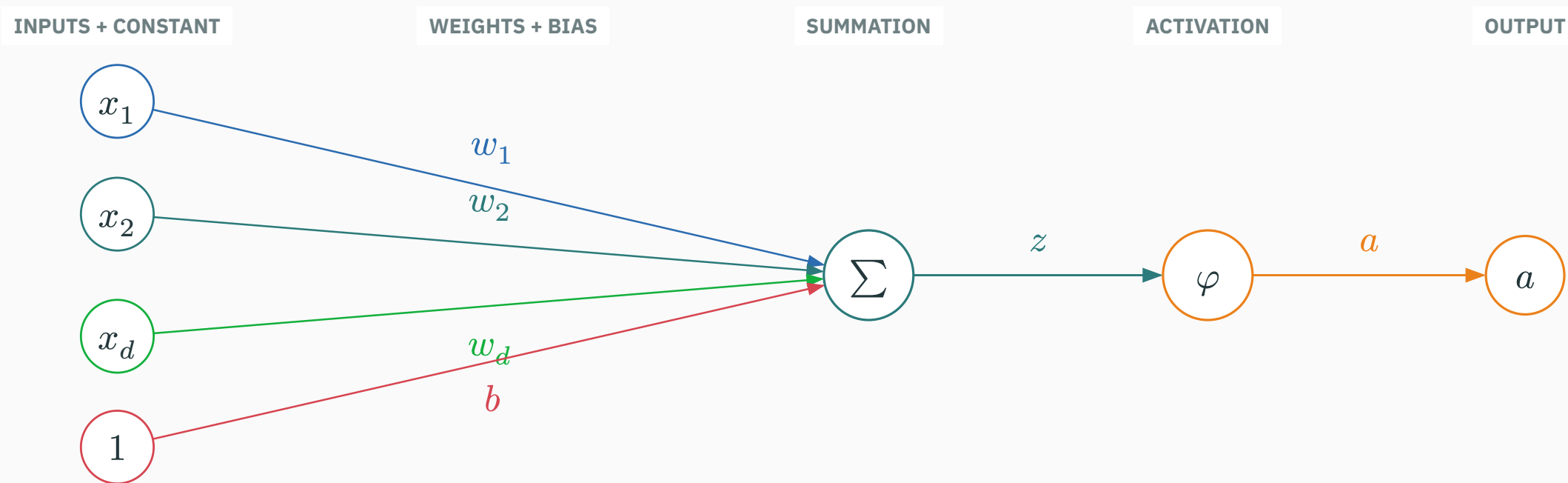


The model can tilt and shift a line or plane, but it cannot bend to follow nonlinear structure.

From a linear score to a neuron

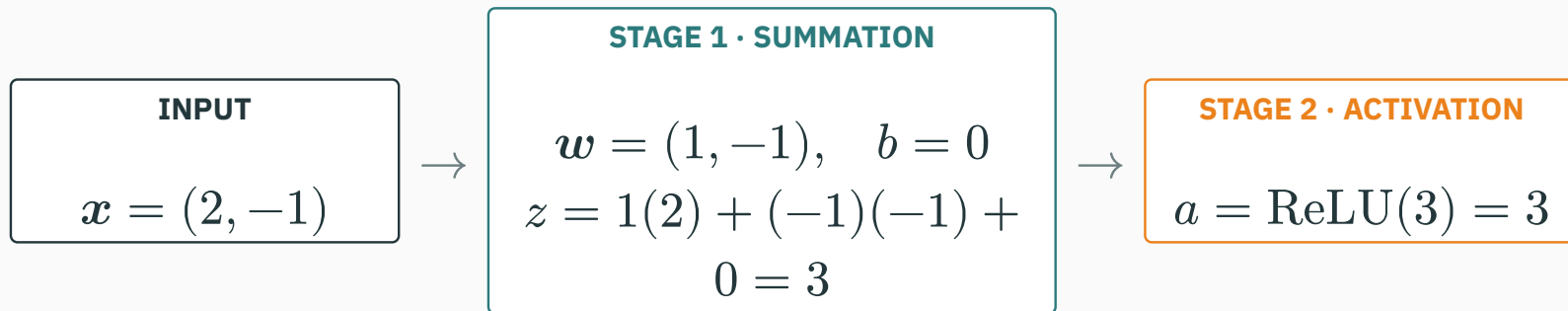
A neuron has two stages

First compute a weighted sum plus bias; then pass that value through an activation.



Weights and bias create z at the Σ node; the separate φ node reshapes z into a .

Trace one concrete neuron through the two stages



Changing w or b changes which input patterns make z large.

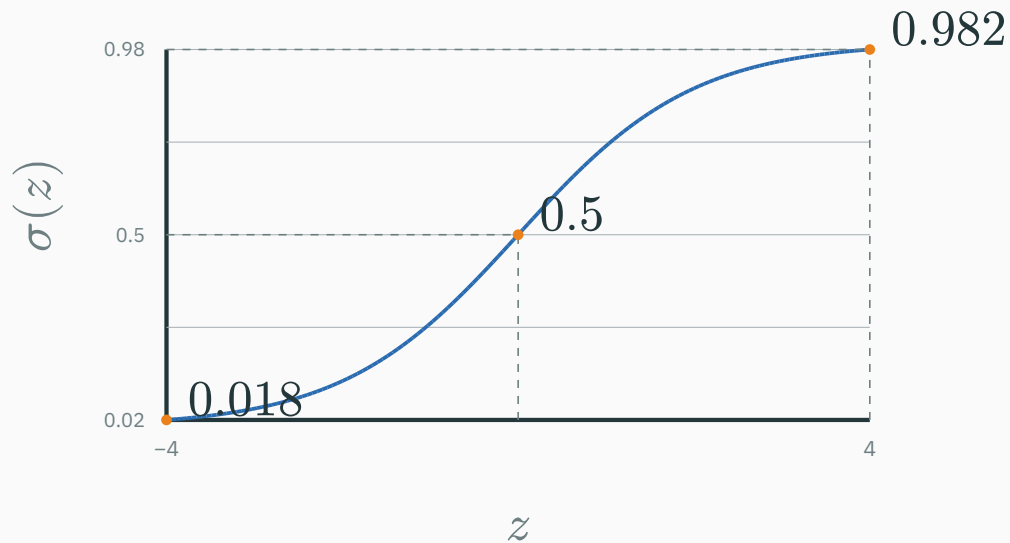
Changing φ changes how the same pre-activation z is gated or reshaped.

Topology tells us where values flow; this calculation tells us the number carried along each stage.

Bounded activations behave like smooth switches

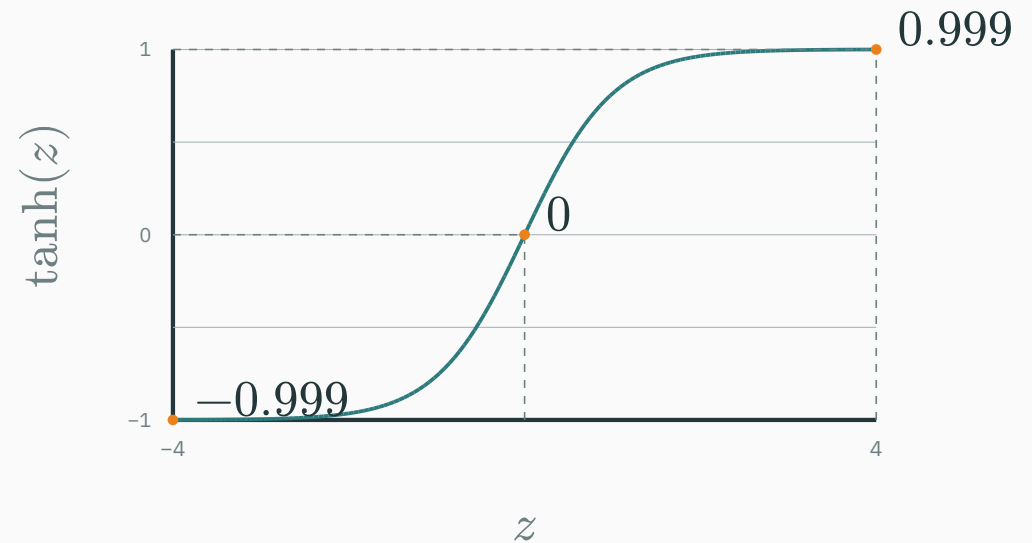
$$\sigma(z) = \frac{1}{1 + \exp(-z)}$$

sigmoid: off $\rightarrow$ on



$$\tanh(z) = \frac{\exp(z) - \exp(-z)}{\exp(z) + \exp(-z)}$$

tanh: negative $\rightarrow$ positive

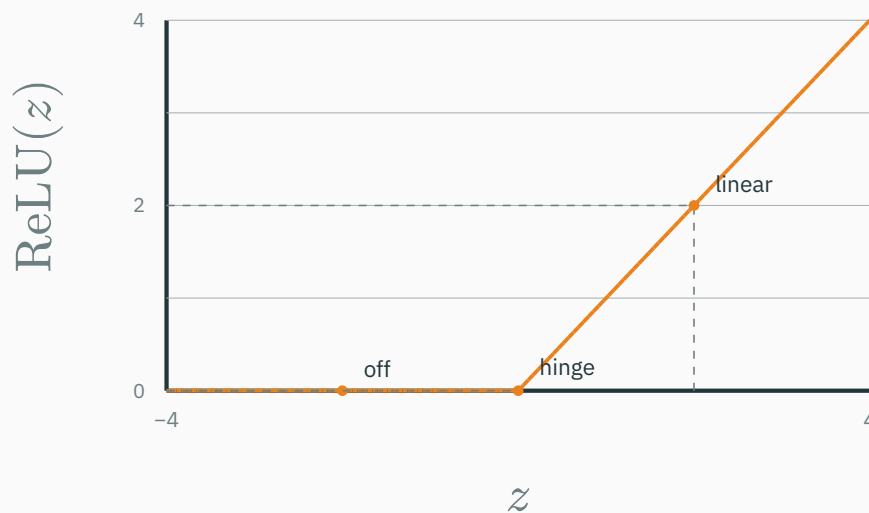


Sigmoid answers “how on?”; tanh answers “negative or positive, and how strongly?”

ReLU and GELU behave like ramps

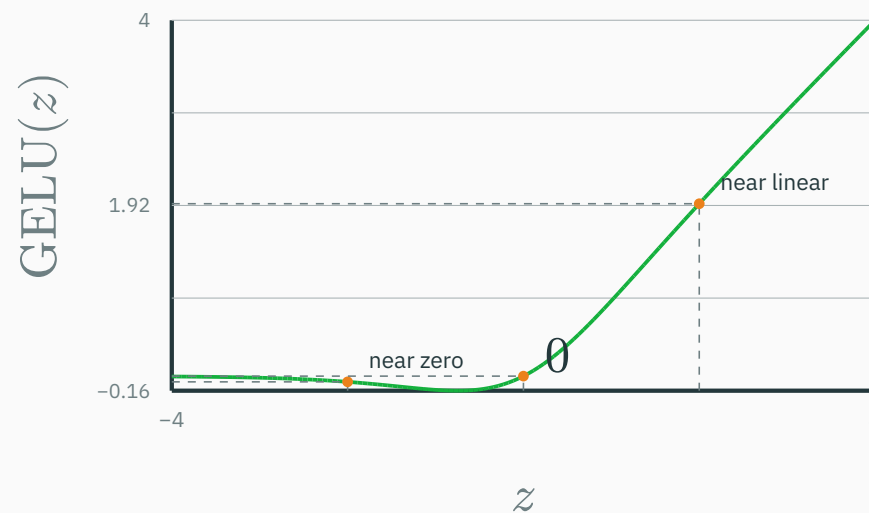
$$\text{ReLU}(z) = \max(0, z)$$

ReLU: hard gate + ramp



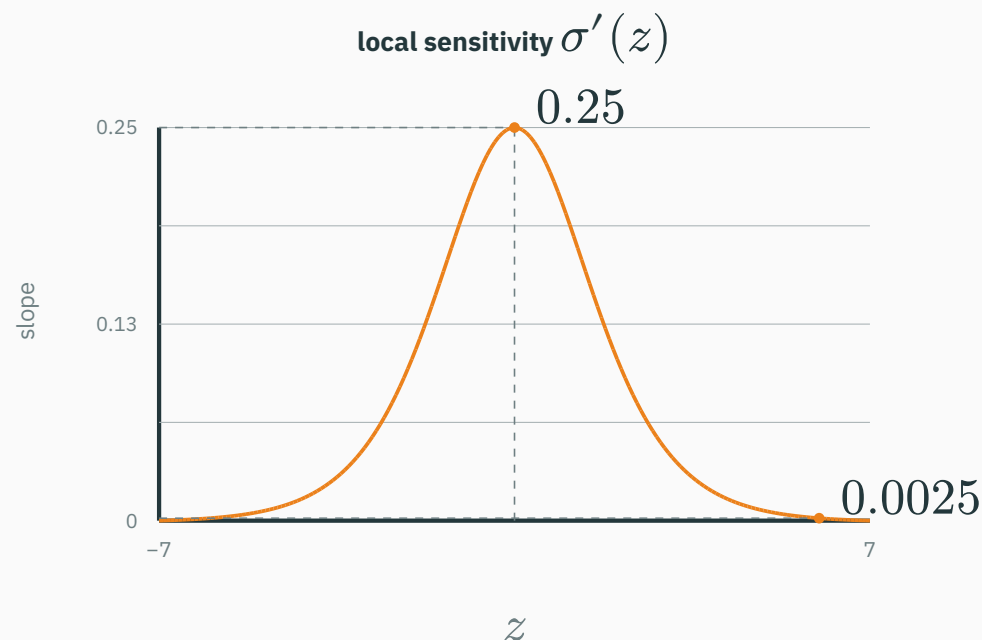
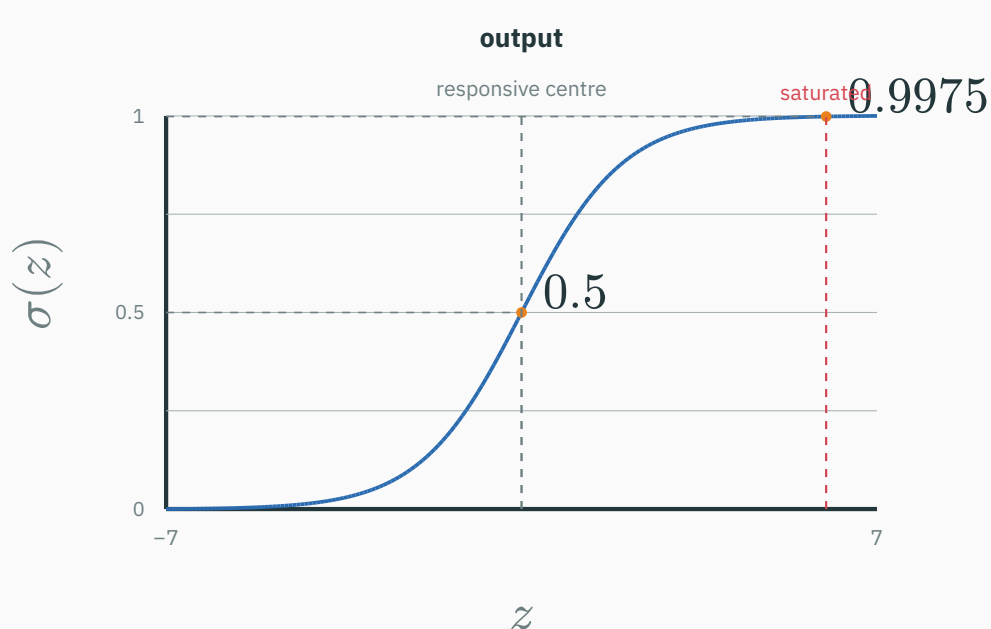
$$\text{GELU}(z) \approx z\sigma(1.702z)$$

GELU: smooth gate + ramp



Both build ramp-like hidden features; ReLU has a sharp hinge, while GELU eases through the gate.

Saturation means the curve has become locally flat



Near $z = 0$, the curve is steep; near $z = 6$, it is almost horizontal.

A saturated unit still outputs a number, but changing its input barely changes that output.

Calculate the effect of saturation

D DERIVATION

CENTRE: $z = 0$

$$\begin{aligned}\sigma'(0) &= 0.25 \\ \text{for } \Delta z &= 0.1: \\ \Delta h &\approx 0.25(0.1) = 0.025\end{aligned}$$

same
 Δz
 $\rightarrow$

SATURATED: $z = 6$

$$\begin{aligned}\sigma'(6) &\approx 0.0025 \\ \text{for } \Delta z &= 0.1: \\ \Delta h &\approx 0.0025(0.1) = \\ &0.00025\end{aligned}$$

$$\frac{0.025}{0.00025} = 100 \rightarrow \text{the saturated unit responds } \mathbf{100\times \text{ less.}}$$

For a positive ReLU input, $\text{ReLU}'(z) = 1$, so the same $\Delta z = 0.1$ gives $\Delta h \approx 0.1$.

The derivative is a local responsiveness meter; Lecture 3 will connect it to the learning signal.

Remove every activation. What can a stack of “weighted sum + bias” layers represent?

A. One equivalent weighted sum + bias

B. Any curved decision boundary

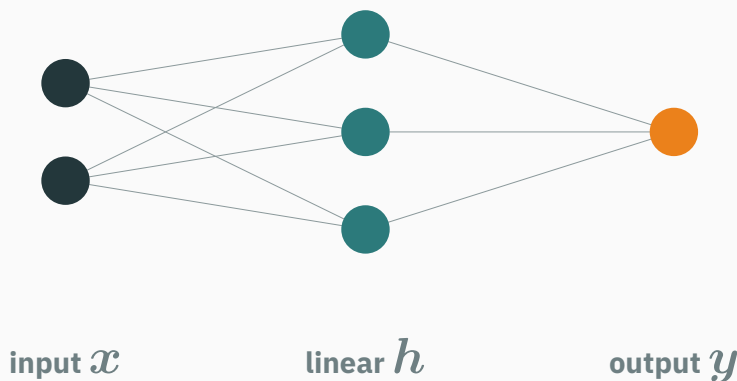
C. Only a constant function

D. A probability distribution automatically

Answer in nodes: two linear layers collapse into one

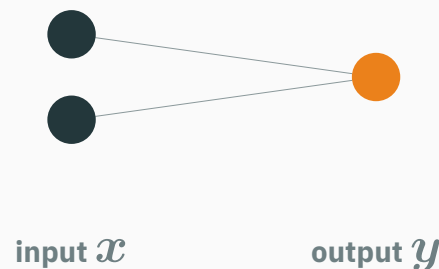
A ANSWER

TWO LINEAR LAYERS



$$h = W_1 x + b_1$$
$$y = W_2 h + b_2$$

ONE EQUIVALENT LAYER



$$y = W_{\text{eq}} x + b_{\text{eq}}$$

no hidden representation is needed

$$W_{\text{eq}} = W_2 W_1, \quad b_{\text{eq}} = W_2 b_1 + b_2$$

Without a φ node, the middle column adds parameters but no new bend or feature shape.

INTERACTIVE

↗ nipunbatra.github.io/dl-teaching/interactives/activation-explorer.html

Compare activations and their **derivatives**, and see where gradients vanish.

Controls: activation type · input z · show derivative · compare saturation.

Explore large positive and negative logits, where sigmoid's derivative becomes tiny.

**Where a single weighted sum
fails**

A binary classifier draws one straight boundary

LINEAR SCORE

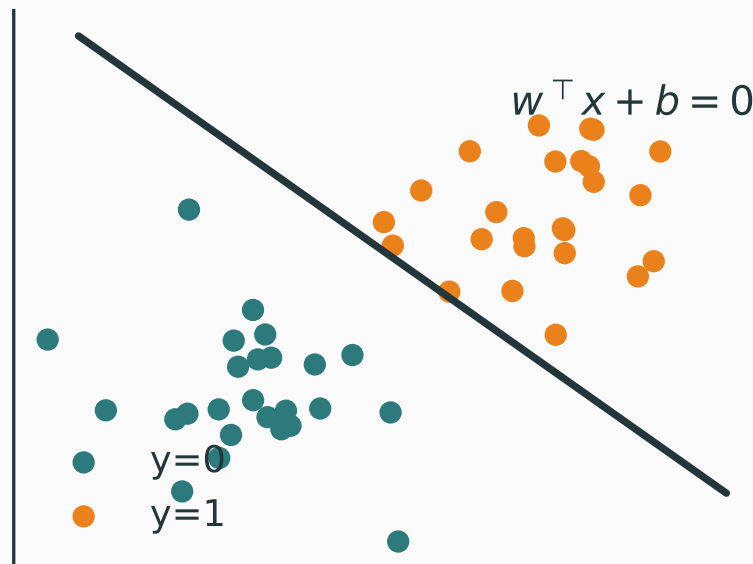
$$z = w^T x + b$$

THRESHOLD

$$\hat{y} = 1 \text{ if } z \geq 0$$

$$\hat{y} = 0 \text{ if } z < 0$$

The boundary is where $z = 0$.



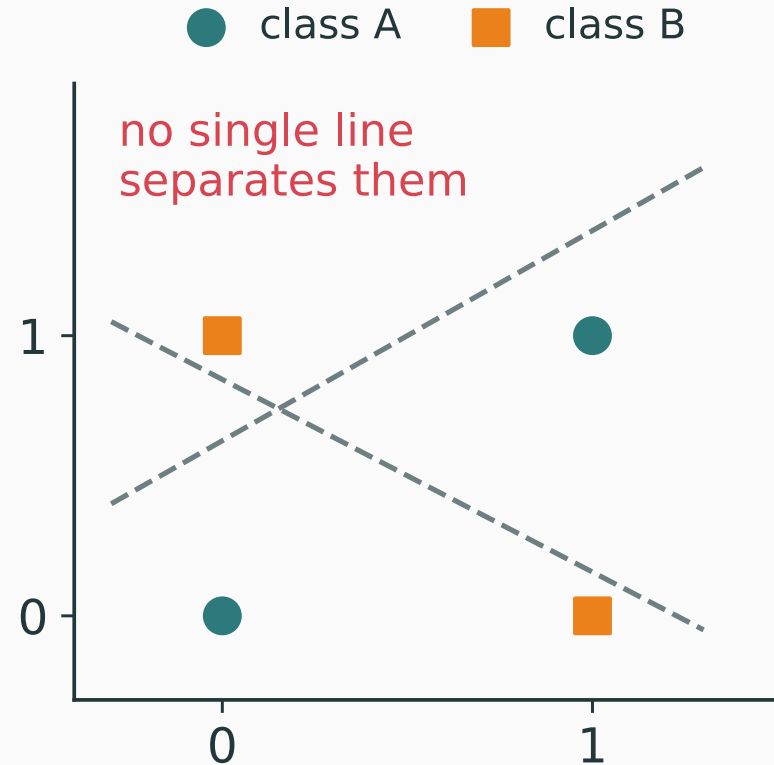
$w^T x + b = 0$ is one line in 2-D, so binary linear classification works only when one line can separate the classes.

XOR is the smallest pattern that defeats one line

CLASS A
 $(0, 0)$ and $(1, 1)$

CLASS B
 $(1, 0)$ and $(0, 1)$

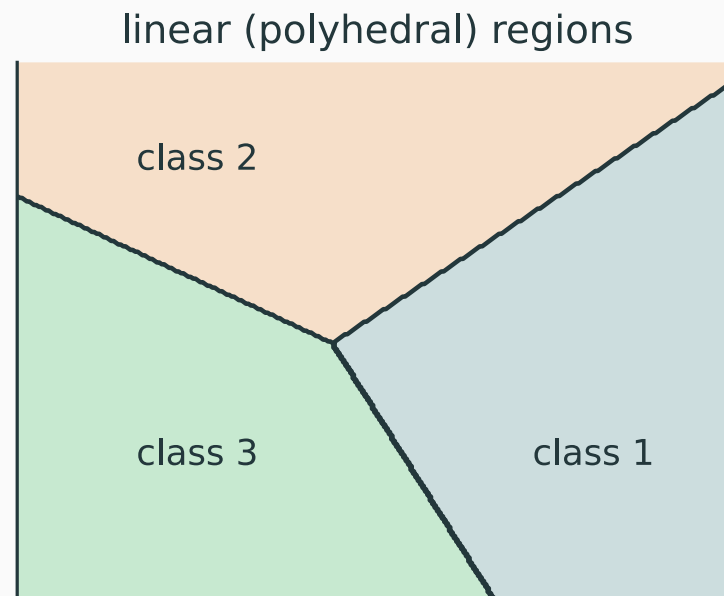
Every proposed line leaves one same-class corner on the wrong side.



XOR requires a nonlinear feature transformation before a final straight separator can work.

For each class k , compute one linear score $z_{k(\mathbf{x})} = \mathbf{w}_k^\top \mathbf{x} + b_k$. Predict the largest score: $\hat{y}(\mathbf{x}) = \arg \max_k z_{k(\mathbf{x})}$.

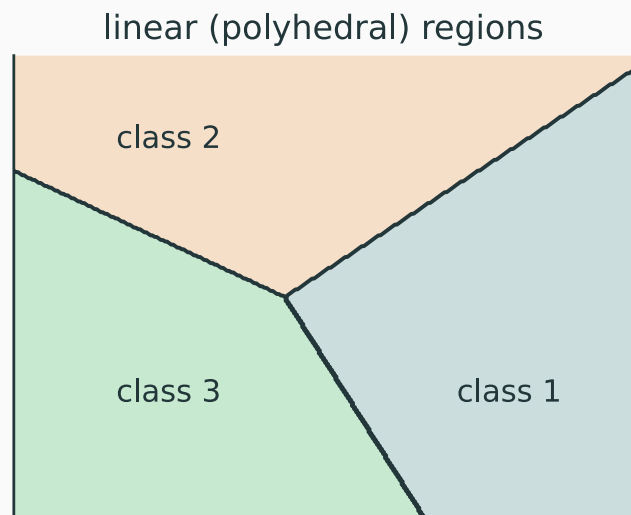
Each pair of classes can exchange the lead only where their two scores tie.



Multi-class regions can form polygons, but every edge still comes from equality between two linear scores.

Question: why are the multi-class boundaries straight?

QUESTION



three linear scores divide the input plane

The classifier predicts

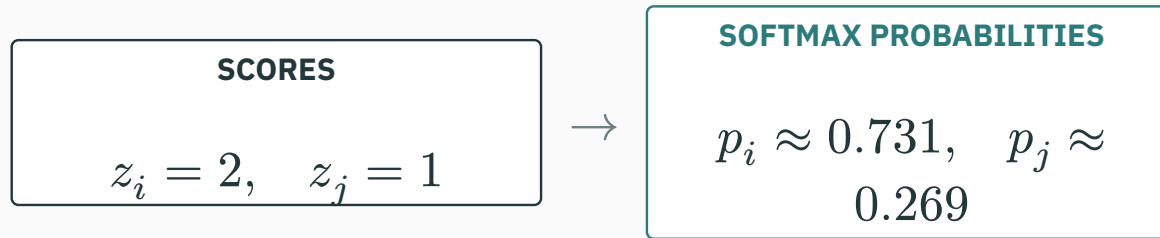
$$\hat{y}(\mathbf{x}) = \arg \max_k p_k(\mathbf{x}).$$

When is a point on the boundary between predicted classes i and j ?

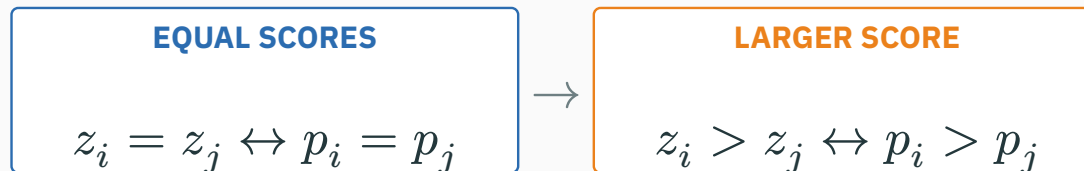
- A.** Whenever $p_i = p_j$
- B.** When $p_i = p_j$ and both are maximal
- C.** Whenever $p_i + p_j = 1$
- D.** Only when all K probabilities are equal

Softmax preserves the winner

A ANSWER



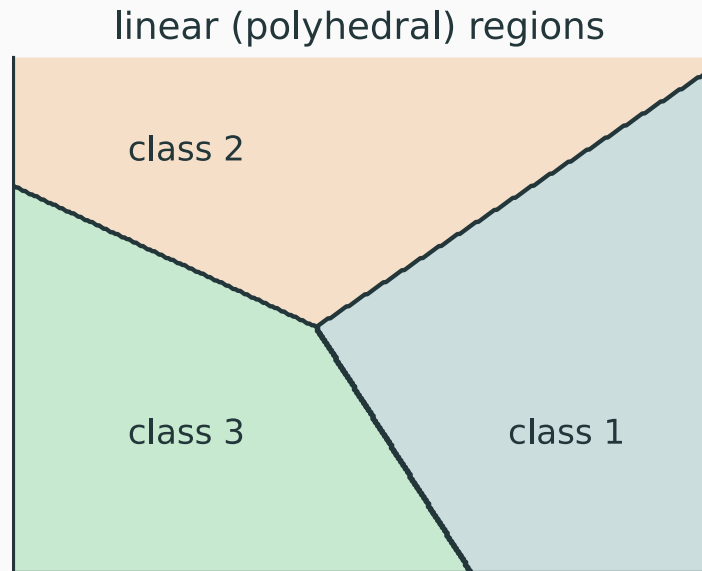
$$\frac{p_i}{p_j} = \frac{\exp(z_i)}{\exp(z_j)} = \exp(z_i - z_j)$$



Softmax changes score scale, not score order: $\arg \max_k p_k = \arg \max_k z_k$.

Linear scores can tie only on a flat set

A ANSWER



For class k , $z_{k(\mathbf{x})} = \mathbf{w}_k^\top \mathbf{x} + b_k$. Two classes tie when $z_{i(\mathbf{x})} = z_{j(\mathbf{x})}$, so $(\mathbf{w}_i - \mathbf{w}_j)^\top \mathbf{x} + (b_i - b_j) = 0$.

This is a line in 2-D, a plane in 3-D, and a hyperplane in d dimensions.

Each coloured region is where one linear score is largest.

Equal linear scores create a candidate straight boundary; softmax cannot bend it.

A concrete example: three linear scores compete

CLASS 1 SCORE

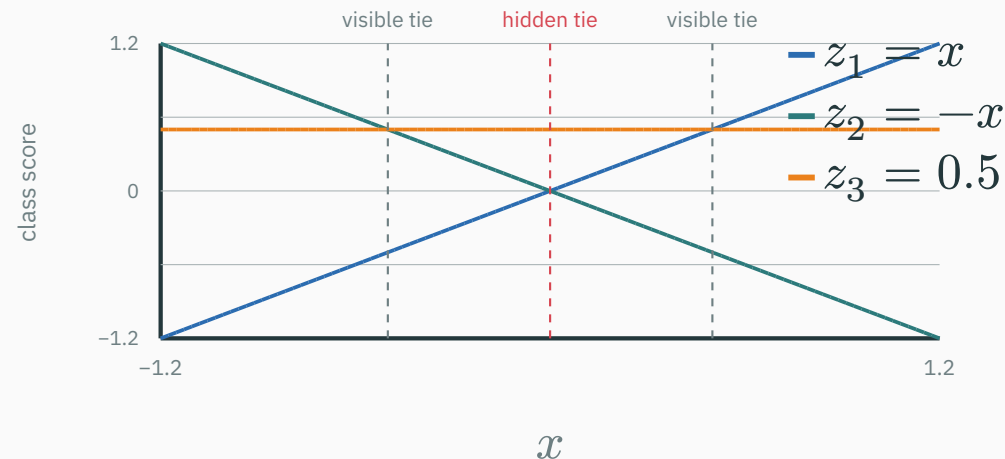
$$z_1(x) = x$$

CLASS 2 SCORE

$$z_2(x) = -x$$

CLASS 3 SCORE

$$z_3(x) = 0.5$$



PREDICTED CLASS = $\arg \max_k z_k(x)$ • boundaries at $x = \pm 0.5$

$$2: x < -.5$$

$$3: |x| < .5$$

$$1: x > .5$$

The learned boundaries are $x = -0.5$ and $x = 0.5$; the class-1/class-2 tie at $x = 0$ remains hidden.

Evaluate five explicit inputs to see which ties are visible

Prediction = $\arg \max_k z_{k(x)}$; a boundary appears only when the tied scores are maximal.

x	$z_1 = x$	$z_2 = -x$	$z_3 = 0.5$	what is visible?
-1	-1	1	0.5	class 2 wins
-0.5	-0.5	0.5	0.5	visible boundary 2/3
0	0	0	0.5	class 3 wins; 1/2 tie hidden
0.5	0.5	-0.5	0.5	visible boundary 1/3
1	1	-1	0.5	class 1 wins

The candidate 1/2 tie occurs at $x = 0$, but the visible boundaries occur at $x = -0.5$ and $x = 0.5$.

Candidate boundary versus visible boundary

A ANSWER

CANDIDATE PAIRWISE TIE

$\mathcal{T}_{ij} = \{x : z_i = z_j\}$
an entire hyperplane

keep
only
winners
→

VISIBLE CLASS BOUNDARY

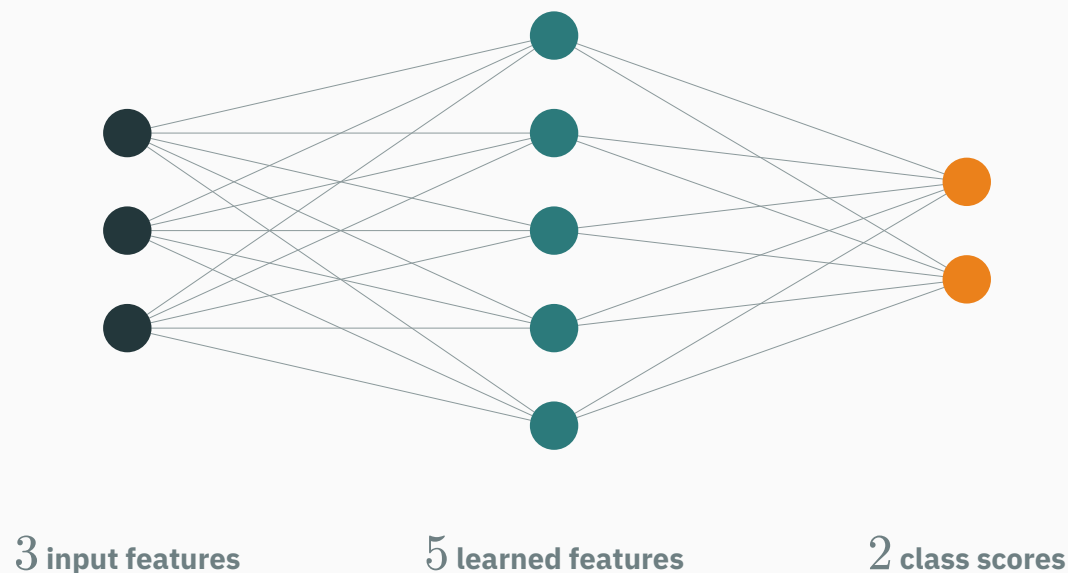
$\mathcal{B}_{ij} = \{x : z_i = z_j \geq z_l \text{ for every } l\}$
the top-scoring portion

In probability language: $p_i = p_j$ **and both are maximal.**

Correct: B. Multi-class linear classifiers have piecewise-straight visible boundaries.

The multilayer perceptron

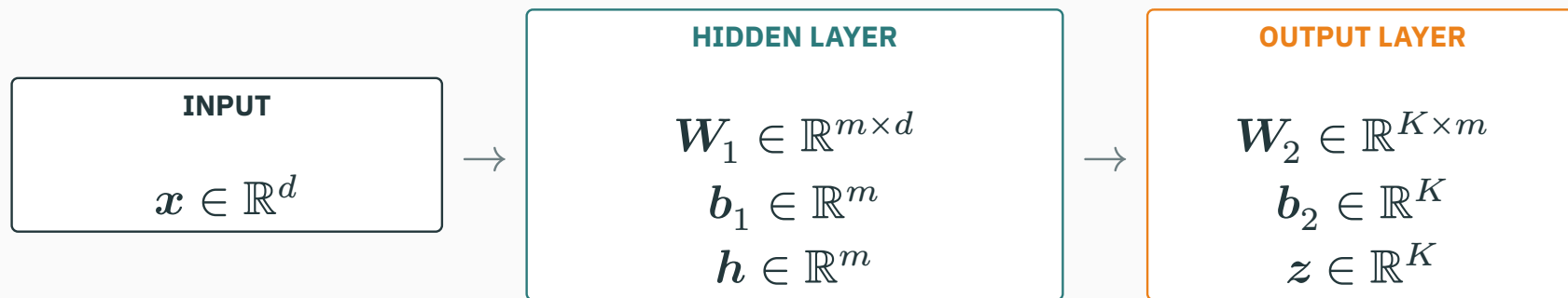
A hidden layer is a new column of learned features



$$h = \varphi(W_1 x + b_1), \quad z = W_2 h + b_2.$$

W_1 stores the $3 \times 5 = 15$ blue-to-teal edges; W_2 stores the $5 \times 2 = 10$ teal-to-orange edges.

General layer dimensions must agree



$$(m \times d)(d \times 1) \rightarrow (m \times 1), \quad (K \times m)(m \times 1) \rightarrow (K \times 1)$$

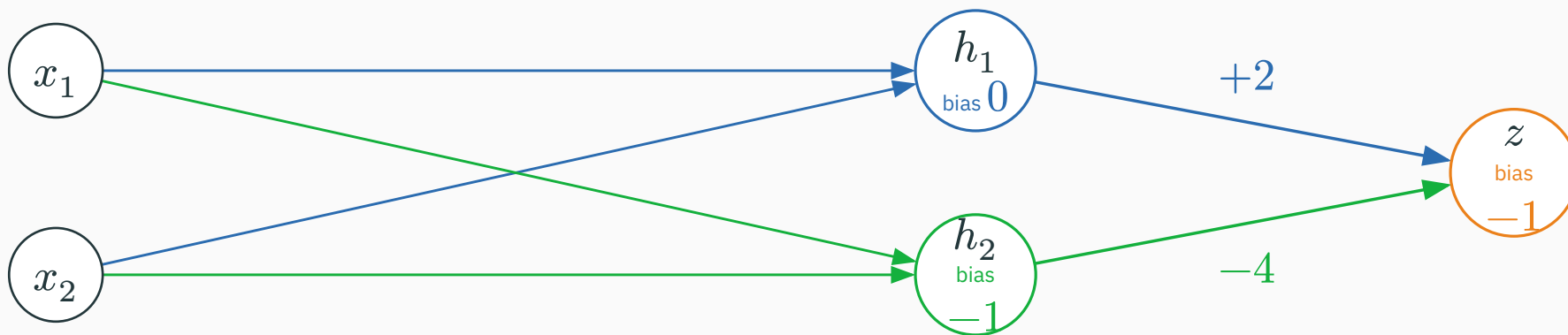
The inner dimensions cancel; the number of rows determines the next representation size.

One XOR network: keep these two features throughout

Both hidden units receive the same pair $x = (x_1, x_2)$:

$$h_1 = \text{ReLU}(x_1 + x_2), \quad h_2 = \text{ReLU}(x_1 + x_2 - 1)$$

all four input weights = $+1$



The output node computes $z = 2h_1 - 4h_2 - 1$; we keep this exact network throughout the XOR example.

Verify the XOR network on all four inputs

Input	$u_1 =$ $x_1 + x_2$	h_1	$u_2 =$ $x_1 +$ $x_2 - 1$	h_2	$z =$ $2h_1 -$ $4h_2 - 1$	$\hat{y}$
(0, 0)	0	0	-1	0	-1	0
(1, 0)	1	1	0	0	1	1
(0, 1)	1	1	0	0	1	1
(1, 1)	2	2	1	1	-1	0

$$\mathbf{W}_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{b}_1 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \mathbf{w}_{\text{out}} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}, \quad b_{\text{out}} = -1.$$

Predict 1 when $z \geq 0$. One ordinary output node correctly classifies all four XOR inputs.

First derive the feature functions for a general input

Let $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$. The weights and bias define a **function** before we substitute any particular input.

BLUE HIDDEN UNIT

$$\mathbf{w}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad b_1 = 0$$

$$\begin{aligned} u_1(\mathbf{x}) &= \mathbf{w}_1^\top \mathbf{x} + b_1 \\ &= 1x_1 + 1x_2 + 0 = x_1 + x_2 \end{aligned}$$

$$\begin{aligned} h_1(\mathbf{x}) \\ &= \text{ReLU}(x_1 + x_2) \end{aligned}$$

GREEN HIDDEN UNIT

$$\mathbf{w}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad b_2 = -1$$

$$\begin{aligned} u_2(\mathbf{x}) &= \mathbf{w}_2^\top \mathbf{x} + b_2 \\ &= 1x_1 + 1x_2 - 1 = x_1 + x_2 - 1 \end{aligned}$$

$$\begin{aligned} h_2(\mathbf{x}) \\ &= \text{ReLU}(x_1 + x_2 - 1) \end{aligned}$$

These are exactly the two XOR features already shown in the node diagram; now we evaluate them at one input.

Check both hidden calculations in matrix form

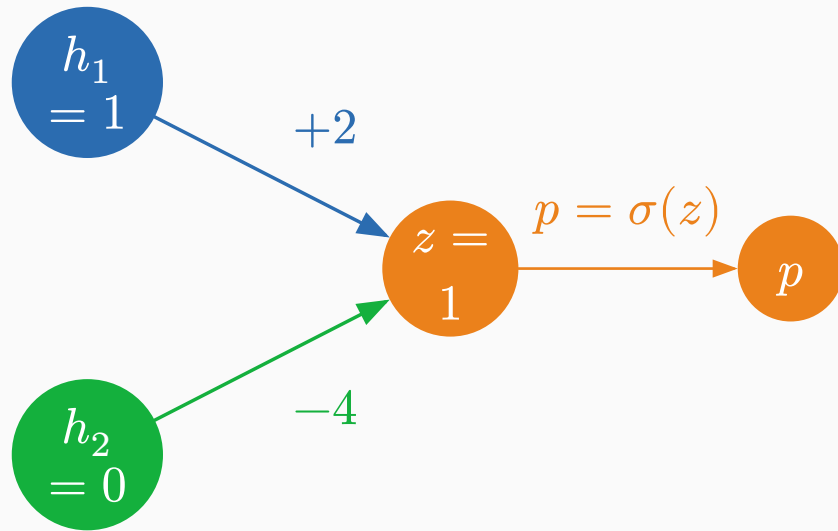
$$\begin{aligned} u &= W_1 x + b_1 \\ &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix}. \end{aligned}$$

$$u_1 = 1(1) + 1(0) + 0 = 1$$

$$u_2 = 1(1) + 1(0) - 1 = 0$$

$$h = \text{ReLU}(u) = \begin{pmatrix} \max(0, 1) \\ \max(0, 0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Rows of W_1 are neurons: each row computes one pre-activation, then ReLU acts elementwise.



Output logit

$$\begin{aligned} z &= (2 \ -4) \begin{pmatrix} 1 \\ 0 \end{pmatrix} - 1 \\ &= 2(1) - 4(0) - 1 = 1. \end{aligned}$$

Probability

$$p = \sigma(1) = \frac{1}{1 + \exp(-1)} \approx 0.731.$$

Binary cross-entropy for $y = 1$

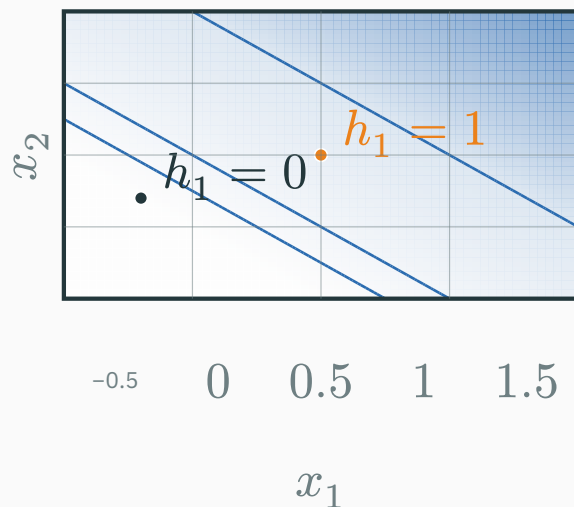
$$\mathcal{L} = -\log(0.731) \approx 0.313.$$

This is the same output node shown in the XOR network: weights $+2, -4$ and bias -1 . Sigmoid only turns z into p .

Each ReLU is flat on one side of a line and rises on the other

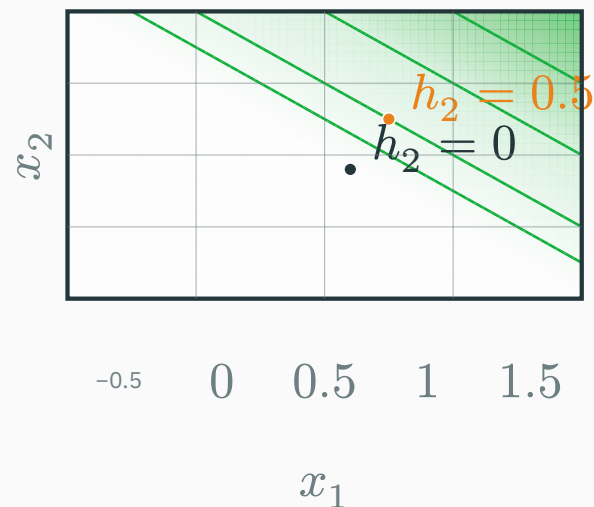
$$h_1(x) = \text{ReLU}(x_1 + x_2)$$

active above $x_1 + x_2 = 0$



$$h_2(x) = \text{ReLU}(x_1 + x_2 - 1)$$

active above $x_1 + x_2 = 1$

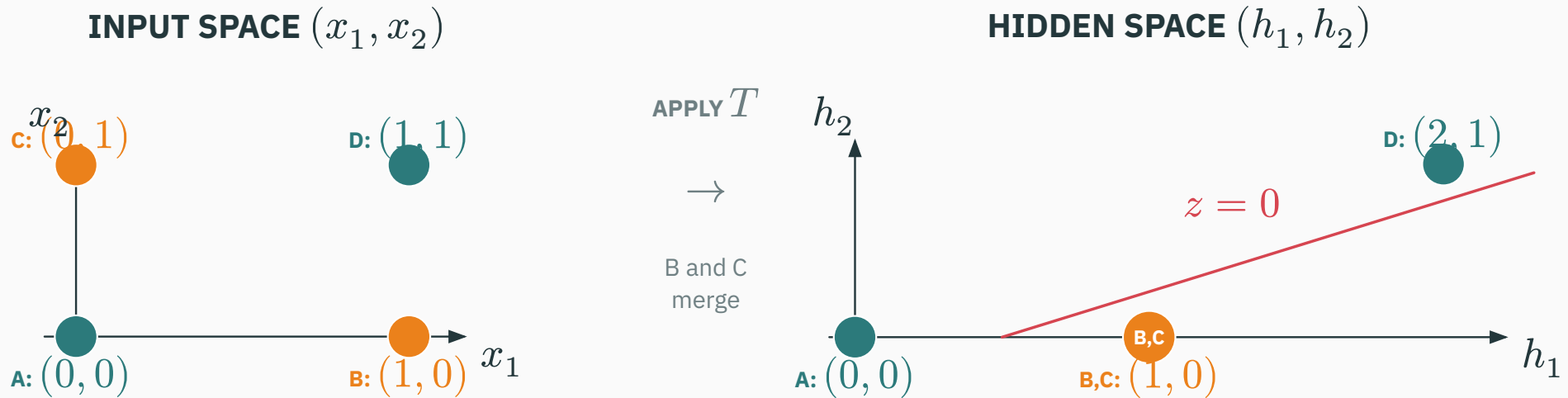


The grid is input space; each colour bar reports a hidden-unit activation value—not a probability.

Compute each XOR point's new hidden-space coordinates

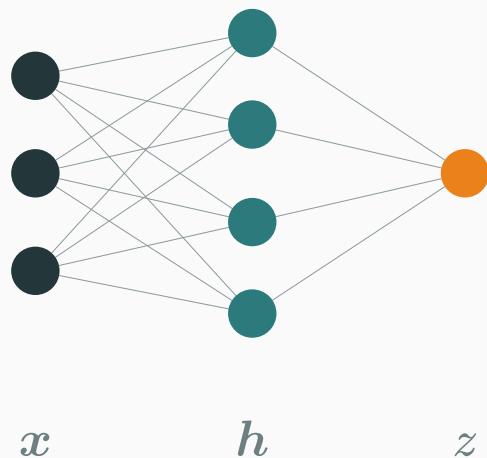
Use the **same map** for every point: $T(x_1, x_2) = (h_1, h_2) = (\text{ReLU}(x_1 + x_2), \text{ReLU}(x_1 + x_2 - 1))$.

A: $(0, 0) \rightarrow (0, 0)$ **B:** $(1, 0) \rightarrow (1, 0)$ **C:** $(0, 1) \rightarrow (1, 0)$ **D:** $(1, 1) \rightarrow (2, 1)$

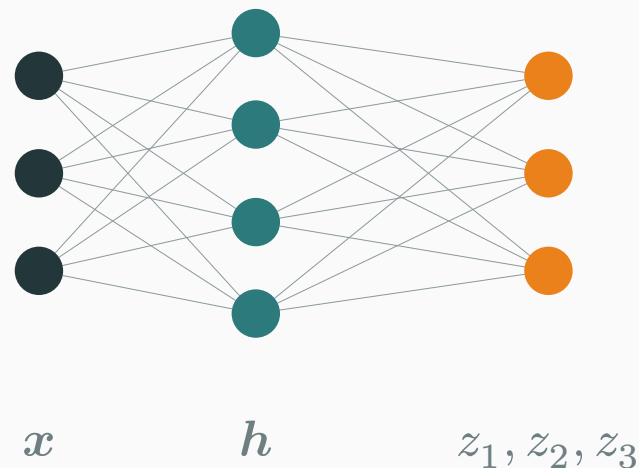


B and C both become $(1, 0)$. Since $z = 2h_1 - 4h_2 - 1$, the boundary $z = 0 \leftrightarrow h_1 - 2h_2 = 0.5$ separates them from A and D.

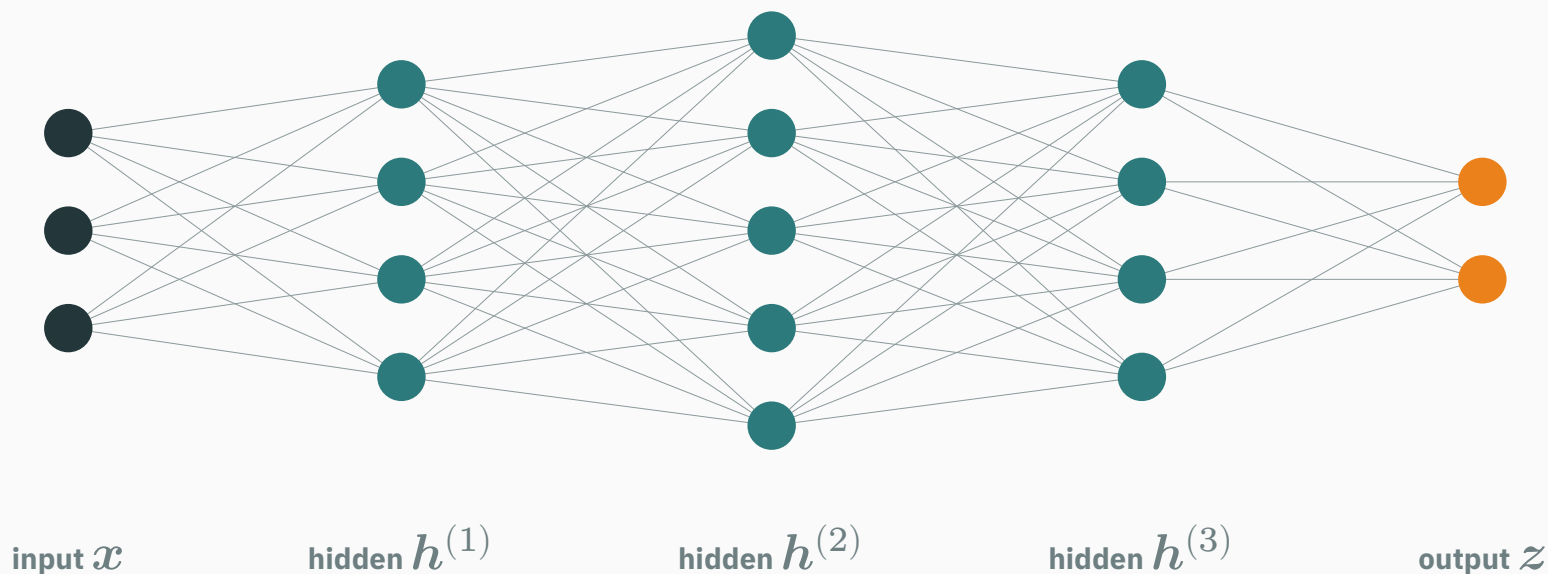
BINARY



THREE CLASSES



The hidden representation can stay the same; only the number and interpretation of output nodes changes.



Every node connects only to the next column; information reaches the output through successive learned transformations.

Depth is visible as the number of hidden columns, not the number of nodes within one column.

Universal approximation

Universal approximation asks one precise question

TARGET REGION TOLERANCE
continuous f $\rightarrow$ closed, bounded D $\rightarrow$ $\varepsilon > 0$

Can some finite network g make $|f(x) - g(x)| < \varepsilon$ for **every** $x \in D$?

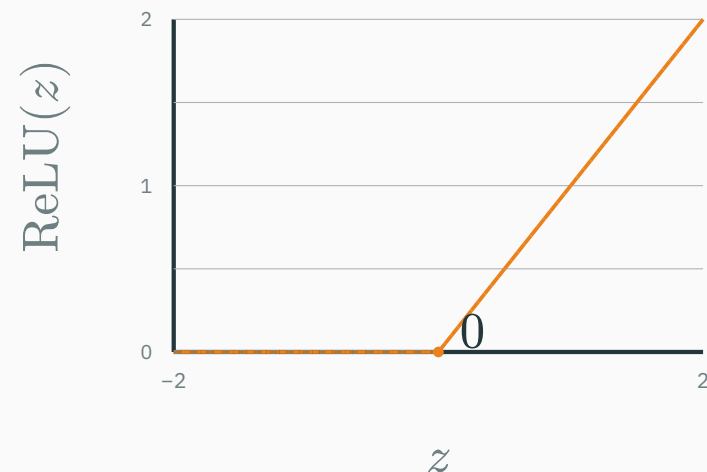
With a suitable nonlinear activation, the theorem says: **yes—such a finite one-hidden-layer network exists.**

This is an existence claim about representational capacity; it does not promise that training will find the network.

A ReLU contributes one hinge

$$\varphi(z) = \max(0, z)$$

- $z < 0$: output is zero;
- $z > 0$: output grows linearly;
- at $z = 0$: the slope changes—a **hinge**.



A hidden layer combines many shifted hinges into a piecewise-linear function.

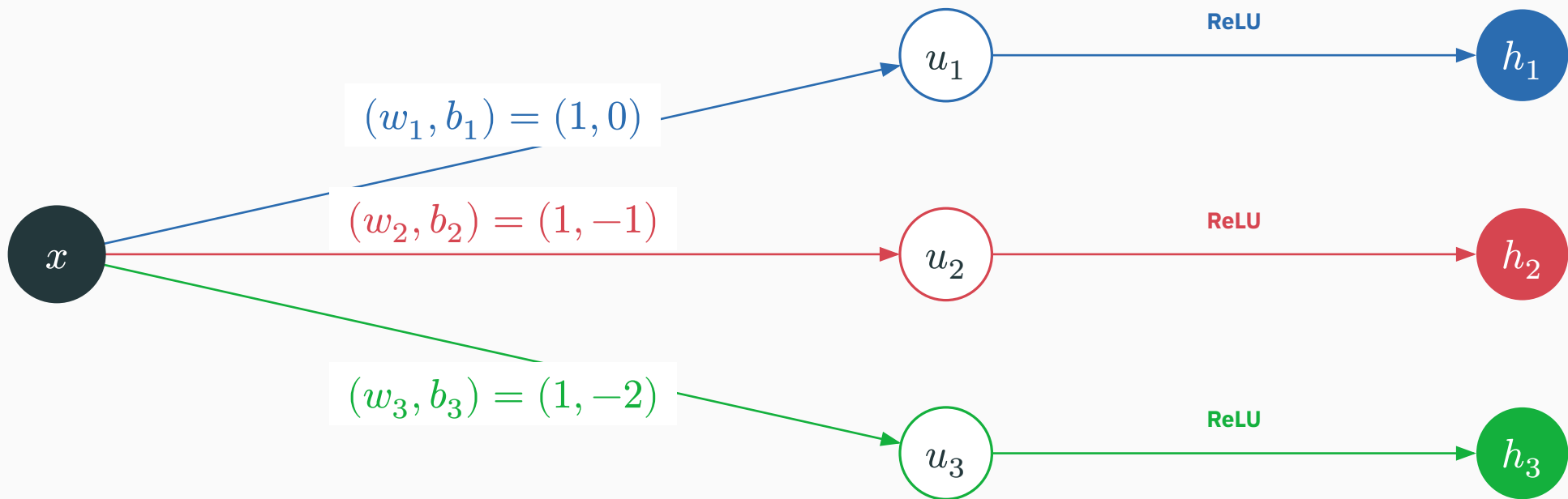
Step 1a: one input fans out to three units

Every unit first computes $u_i = w_i x + b_i$, then applies $h_i = \text{ReLU}(u_i)$.

ONE INPUT

AFFINE SUMS

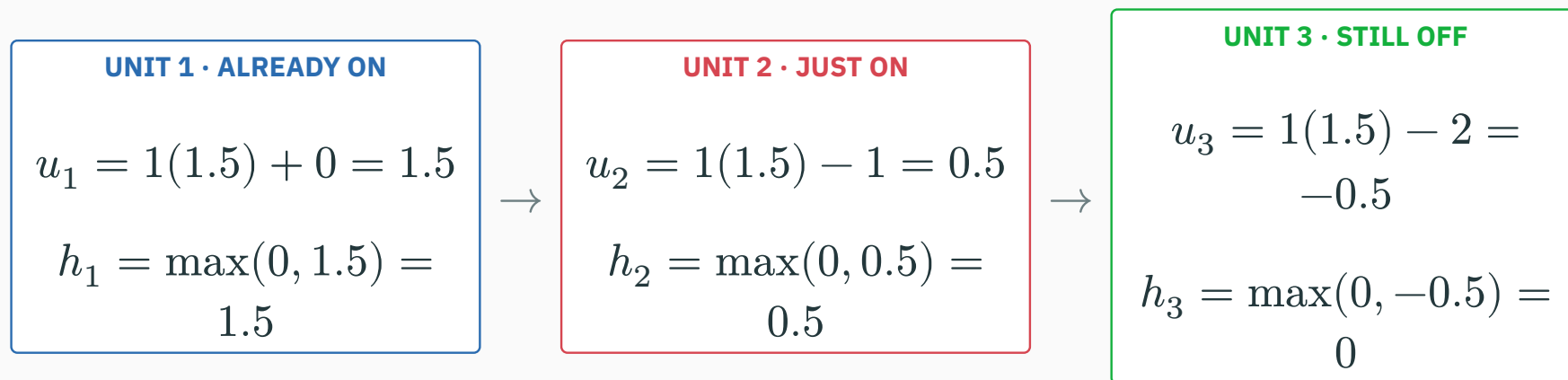
RELU FEATURES



One x fans out; biases $0, -1, -2$ shift the three ReLU thresholds.

Step 1b: evaluate a concrete input

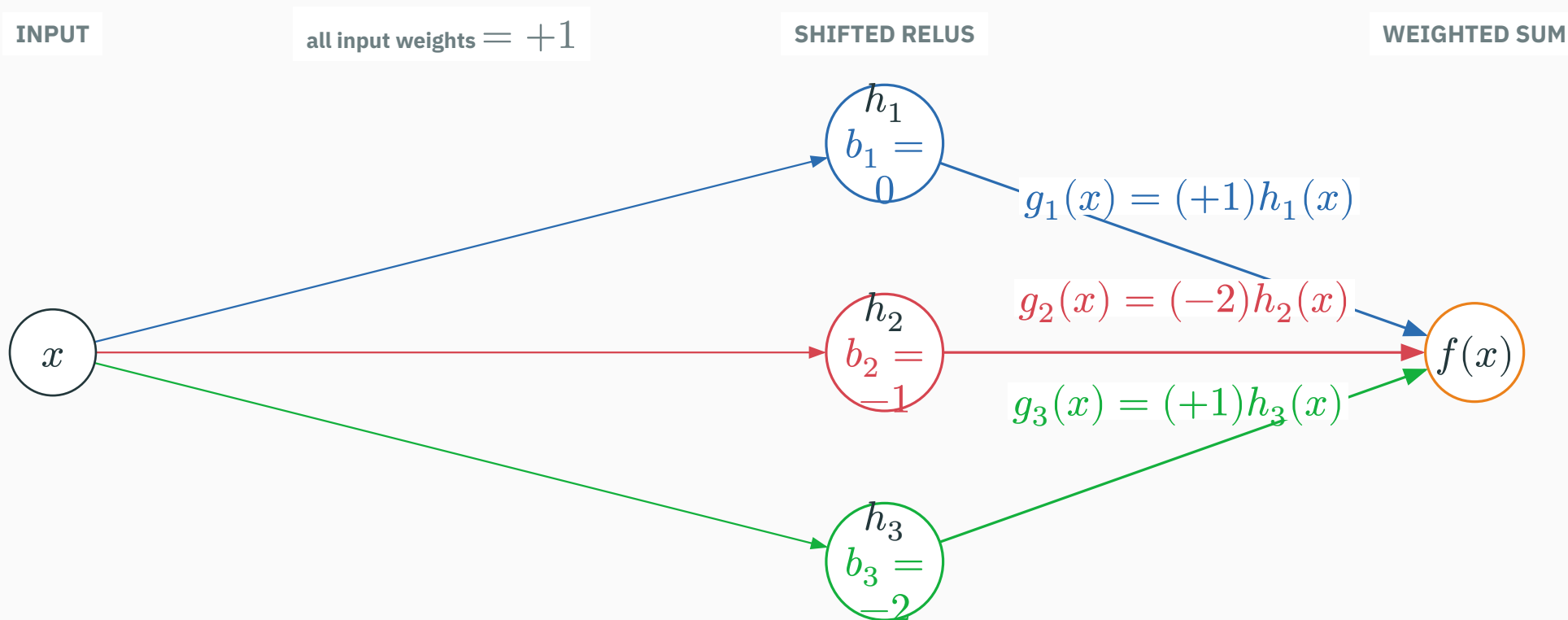
Substitute the same value into all three affine sums, then apply ReLU separately.



$$\mathbf{u} = \begin{pmatrix} 1.5 \\ 0.5 \\ -0.5 \end{pmatrix} \rightarrow \mathbf{h} = \text{ReLU}(\mathbf{u}) = \begin{pmatrix} 1.5 \\ 0.5 \\ 0 \end{pmatrix}$$

At $x = 1.5$, units 1 and 2 are active; unit 3 stays zero until x passes 2.

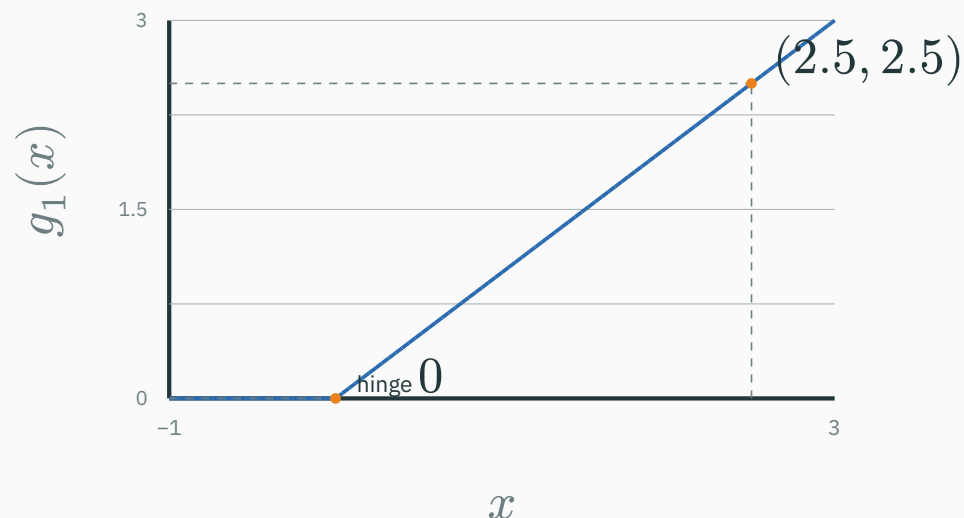
Step 2: output edges combine the three learned hinges



The output adds $f(x) = g_1(x) + g_2(x) + g_3(x)$ —the theorem’s “sum of hidden units” is literal.

Step 3a: the positive output weights keep the ramps upright

$$g_1(x) = (+1)h_1$$



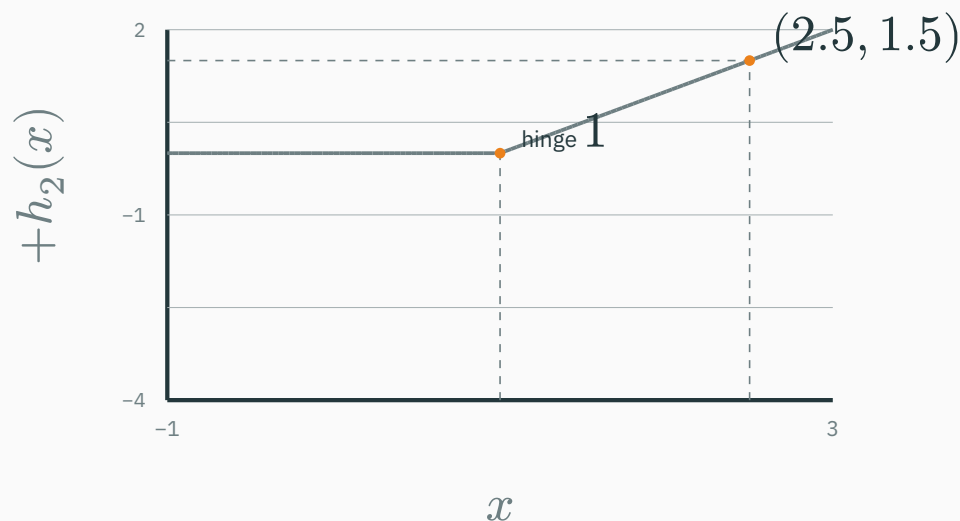
$$g_3(x) = (+1)h_3$$



A positive output weight preserves the direction of a hidden ReLU ramp.

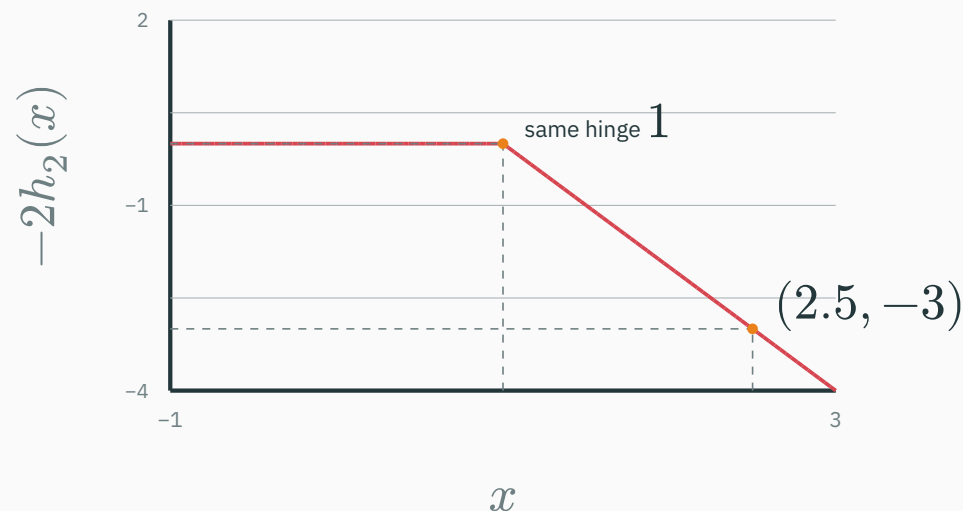
Step 3b: a negative weight flips the ramp

$$(+1)h_2(x) = \text{ReLU}(x - 1)$$



slope $+1$ after the hinge

$$(-2)h_2(x) = -2\text{ReLU}(x - 1)$$

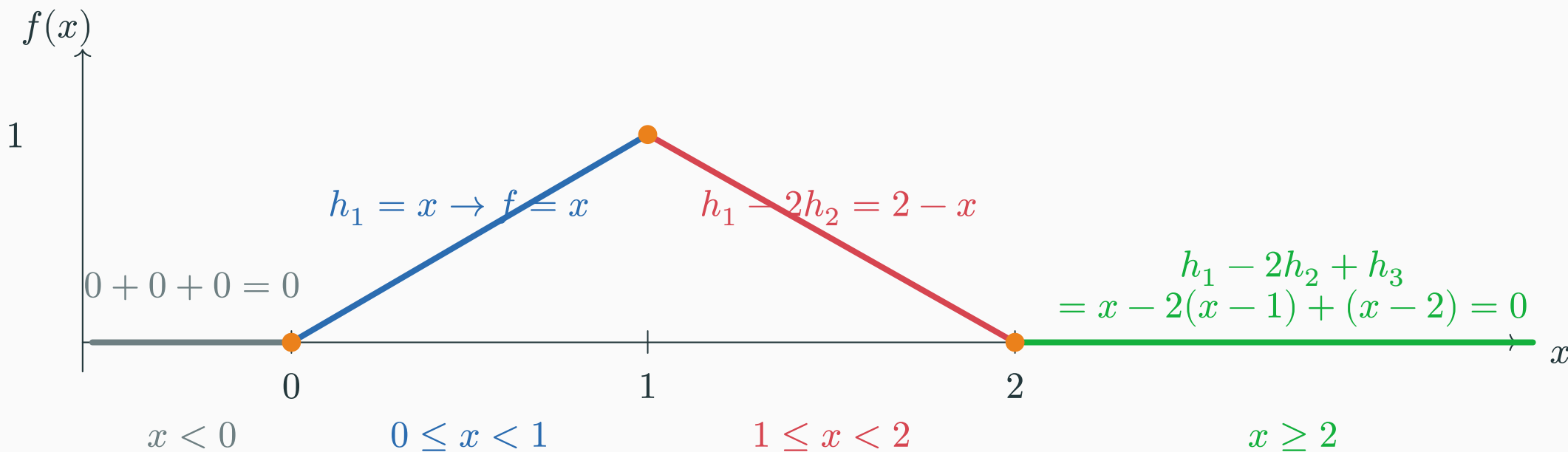


slope -2 after the hinge

Multiplying by -2 keeps the hinge at $x = 1$, flips the ramp, and doubles its slope magnitude; h_2 itself is unchanged.

Step 4: assemble the tent interval by interval

$$f(x) = h_1 - 2h_2 + h_3 = \text{ReLU}(x) - 2\text{ReLU}(x - 1) + \text{ReLU}(x - 2).$$



Region 4: h_3 switches on and cancels the remaining slope, so $f(x) = 0$.

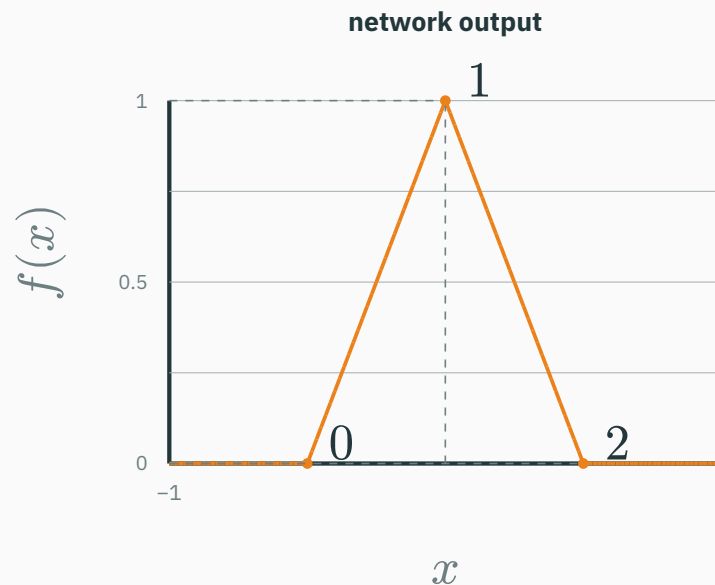
Reading left to right: flat $\rightarrow$ rise $\rightarrow$ fall $\rightarrow$ flat—the weighted sum is a tent.

Step 5: the weighted sum is the tent

Hidden layer $h(x) =$
 $(\text{ReLU}(x), \text{ReLU}(x - 1), \text{ReLU}(x - 2)).$

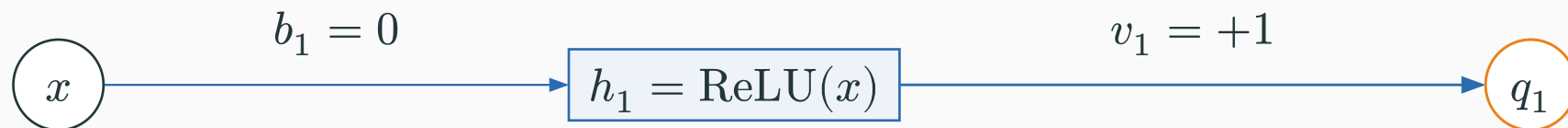
Linear output layer $f(x) =$
 $(1 \ -2 \ 1)h(x).$

The network has **three hidden nodes** and **one output node**. The output node performs no ReLU here; it only takes a weighted sum.



Hidden units create reusable nonlinear features; the output layer chooses their signs and strengths.

Hinge 1 starts a rise

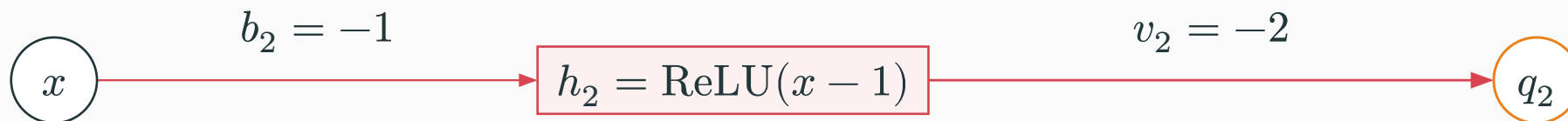


$$q_1 = h_1 \quad | \quad s : 0 + (+1) = +1$$

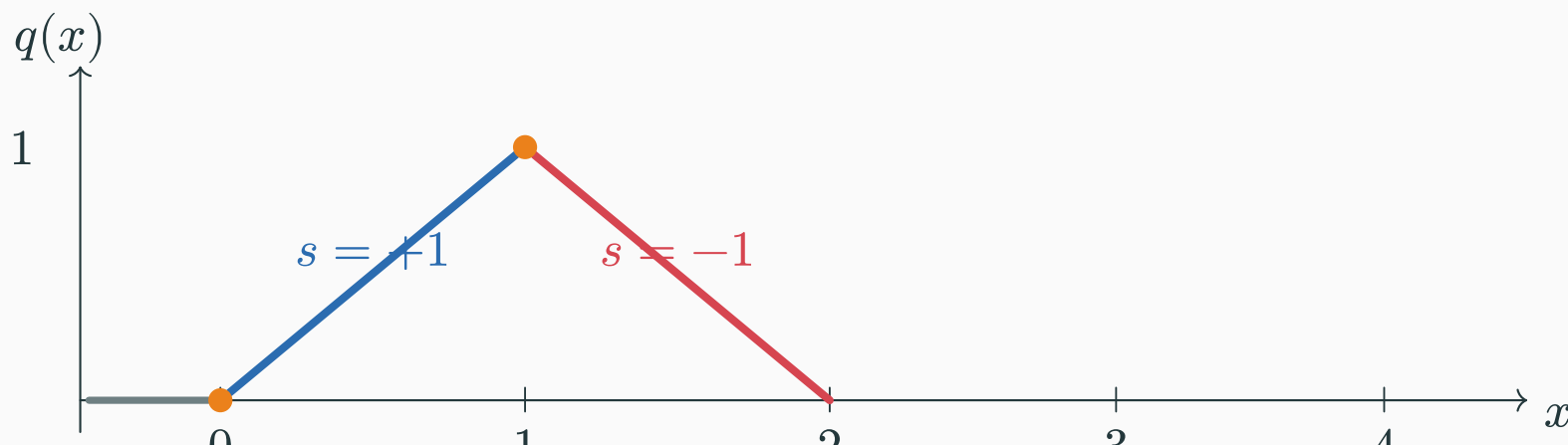


At $x = 0$, the first hidden unit turns on and the output begins rising.

Hinge 2 turns downward

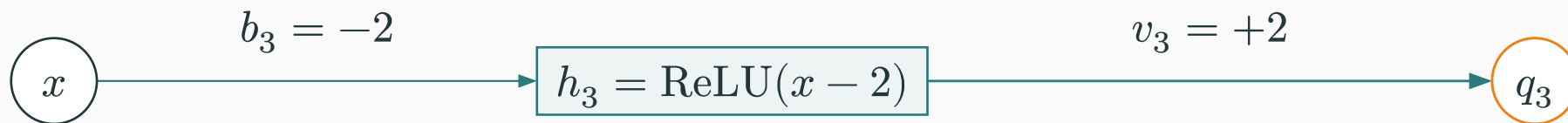


$$q_2 = q_1 - 2h_2 \quad | \quad s : +1 - 2 = -1$$

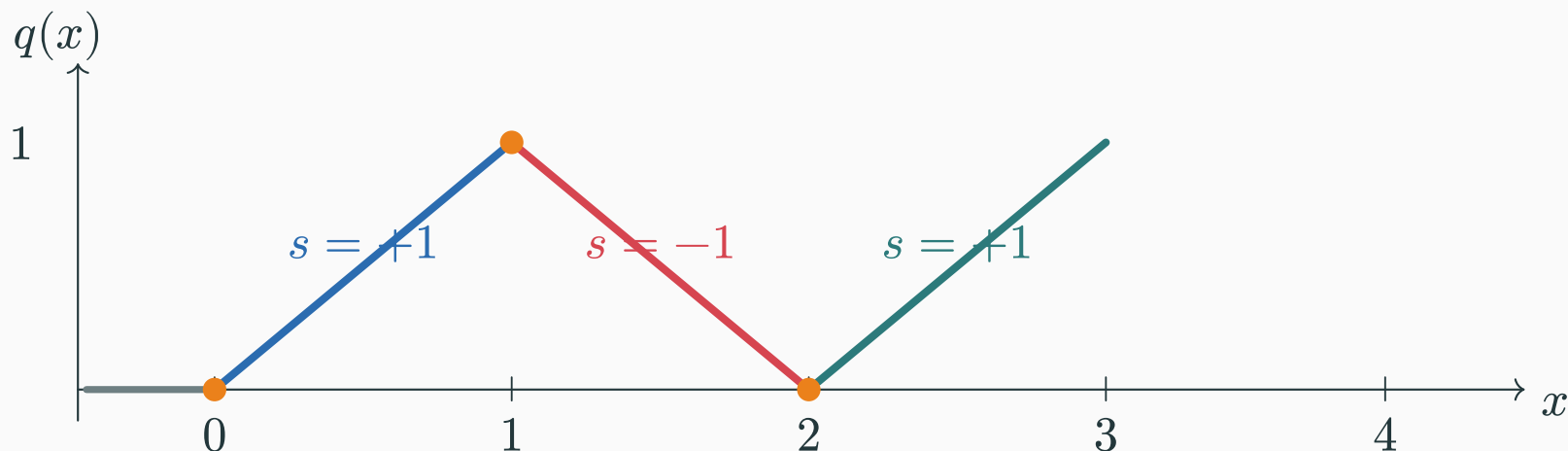


At $x = 1$, the weight -2 changes the slope by -2 : the first peak turns downward.

Hinge 3 turns upward

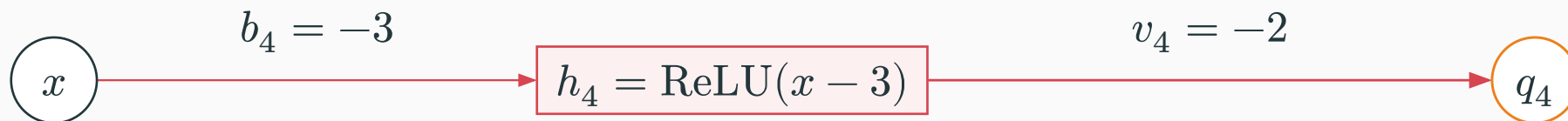


$$q_3 = q_2 + 2h_3 \quad | \quad s : -1 + 2 = +1$$

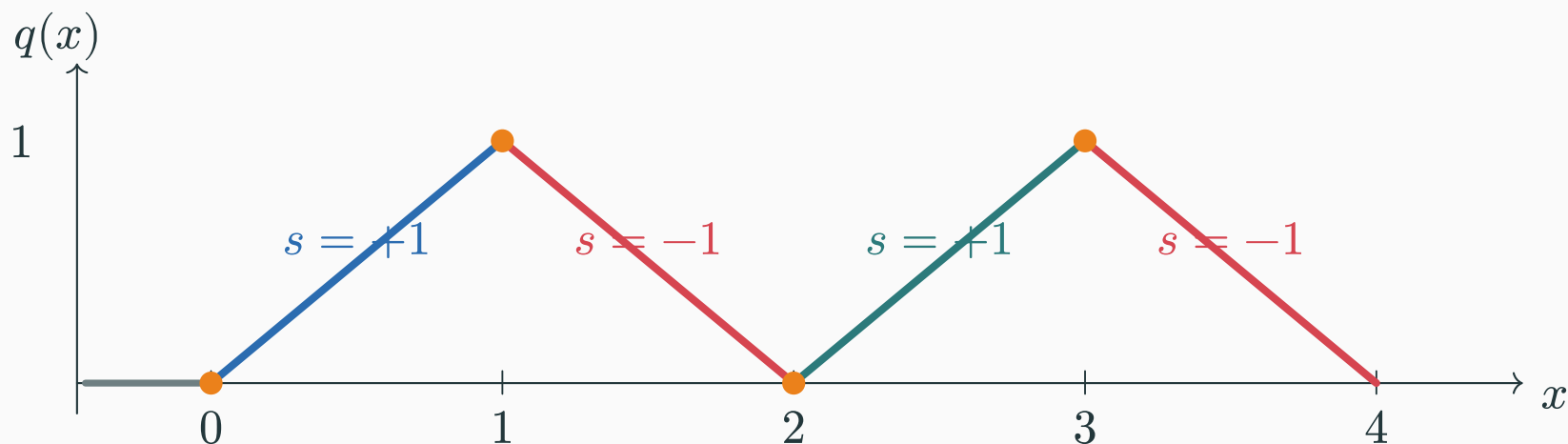


At $x = 2$, the weight $+2$ changes the slope by $+2$: the output rises again.

Hinge 4 turns downward

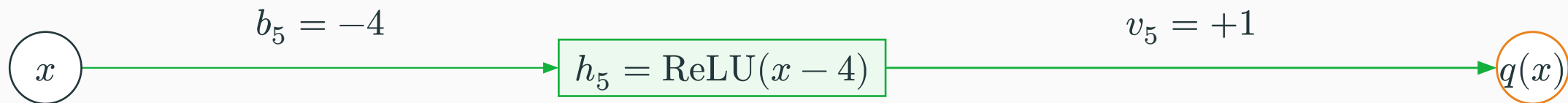


$$q_4 = q_3 - 2h_4 \quad | \quad s : +1 - 2 = -1$$

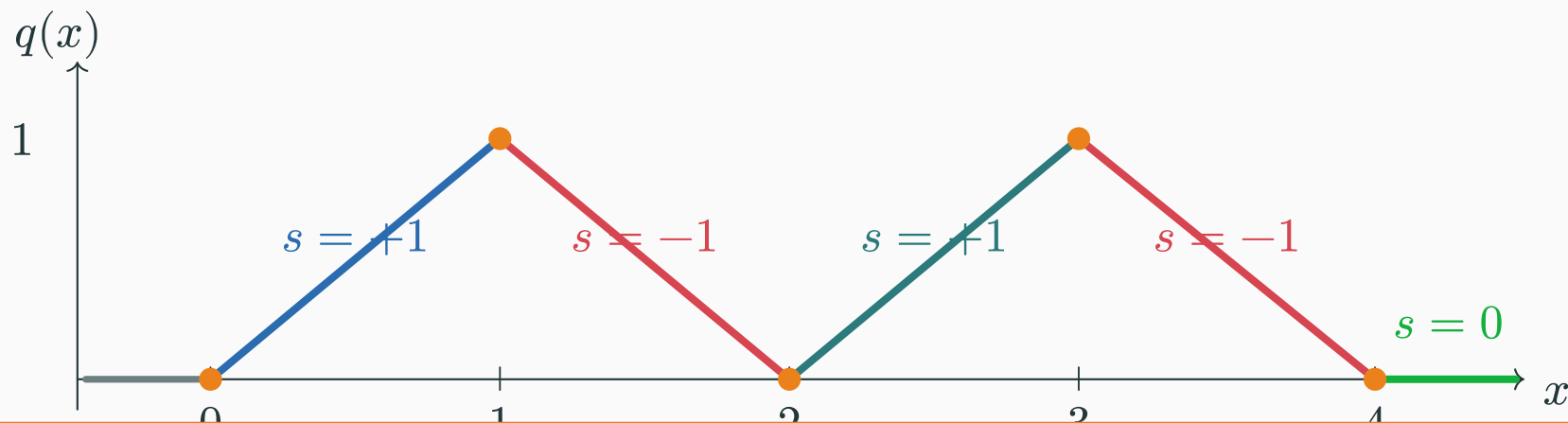


At $x = 3$, another weight -2 turns the second peak downward.

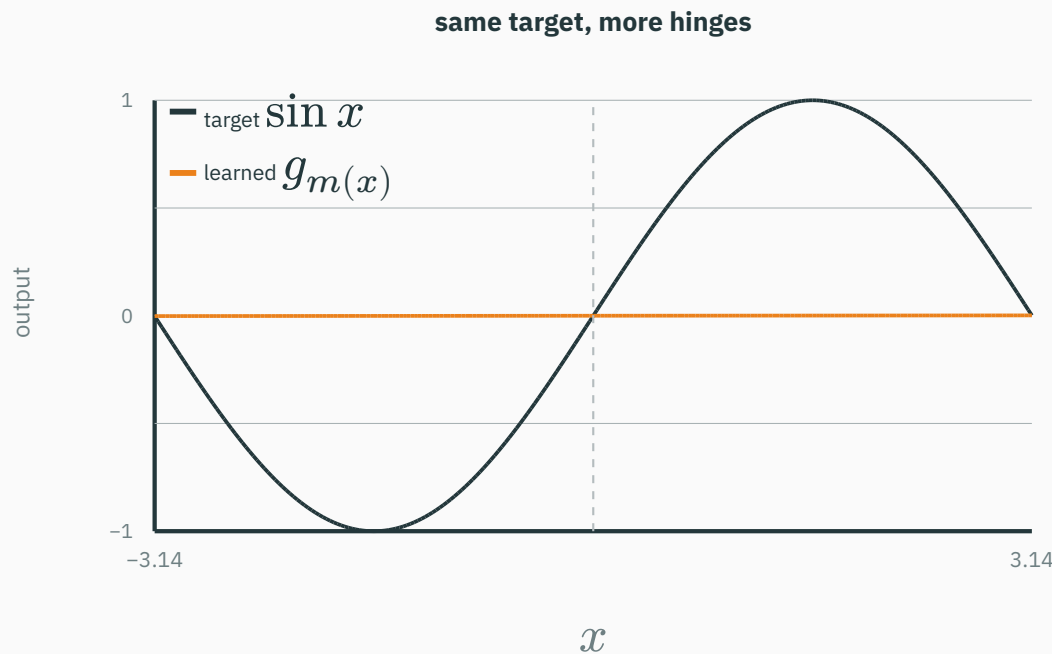
Hinge 5 makes the output flat



$$q = h_1 - 2h_2 + 2h_3 - 2h_4 + h_5 \quad | \quad s : -1 + 1 = 0$$



Five hidden units create five slope changes—and therefore a two-peak piecewise-linear function.



LEARNED FUNCTION

Here: **1 interior breakpoints.**

$$t_i = -\pi + \frac{2\pi i}{2}, \quad i = 0, \dots, 2$$

$$g_m(t_i) = \sin(t_i)$$

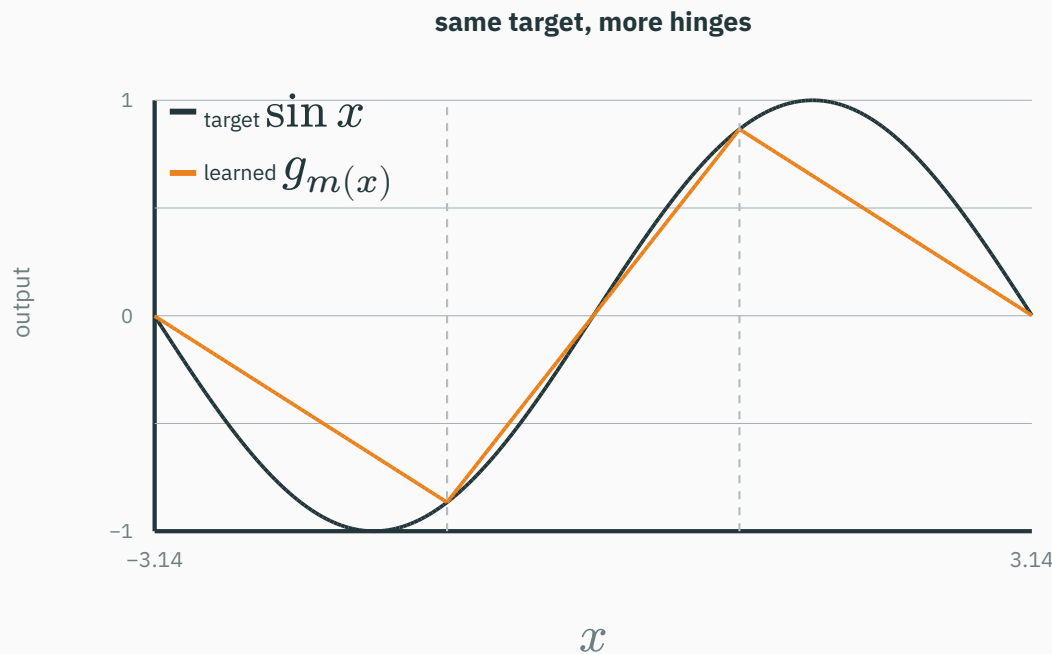
Linear between adjacent knots.

$$\text{ReLU form on } [-\pi, \pi] \quad g_m(x) = a + bx + \sum_{j=1}^m c_j \text{ReLU}(x - t_j)$$

$$\text{Worst vertical gap } \varepsilon_m = \sup_x |\sin x - g_m(x)| \approx 1$$

At knots $-\pi, 0, \pi$, the target is zero, so $g_1(x) = 0$: one hinge cannot capture both lobes.

2 hinges: the curve begins to bend



LEARNED FUNCTION

Here: **2 interior breakpoints**.

$$t_i = -\pi + \frac{2\pi i}{3}, \quad i = 0, \dots, 3$$

$$g_m(t_i) = \sin(t_i)$$

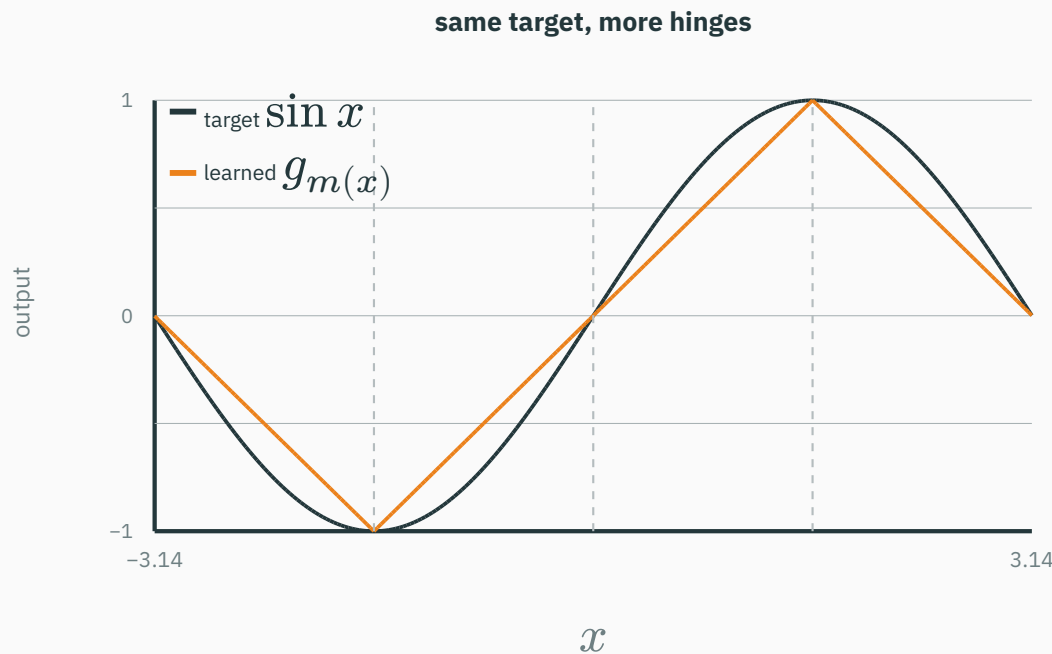
Linear between adjacent knots.

$$\text{ReLU form on } [-\pi, \pi] \quad g_m(x) = a + bx + \sum_{j=1}^m c_j \text{ReLU}(x - t_j)$$

$$\text{Worst vertical gap } \varepsilon_m = \sup_x |\sin x - g_m(x)| \approx 0.437$$

The learned function now captures the sign change, but its peaks remain coarse.

3 hinges: the sine shape appears



LEARNED FUNCTION

Here: **3 interior breakpoints**.

$$t_i = -\pi + \frac{2\pi i}{4}, \quad i = 0, \dots, 4$$

$$g_m(t_i) = \sin(t_i)$$

Linear between adjacent knots.

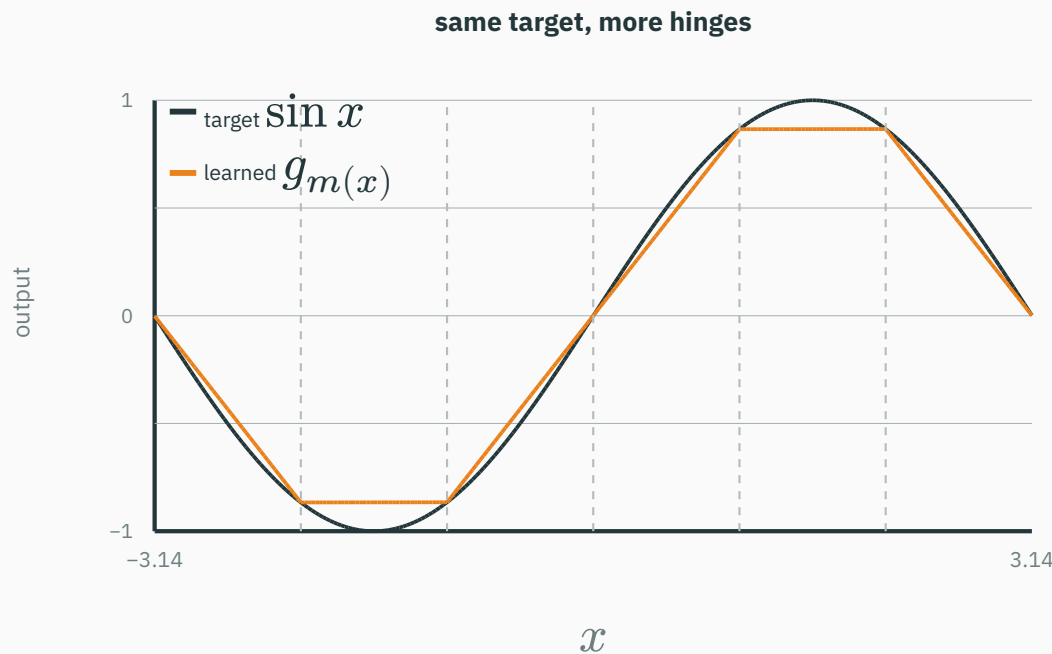
ReLU form on $[-\pi, \pi]$ $g_m(x) =$

$$a + bx + \sum_{j=1}^m c_j \text{ReLU}(x - t_j)$$

Worst vertical gap $\varepsilon_m =$

$$\sup_x |\sin x - g_m(x)| \approx 0.211$$

Three hinges capture both extrema and the zero crossing.



LEARNED FUNCTION

Here: **5 interior breakpoints**.

$$t_i = -\pi + \frac{2\pi i}{6}, \quad i = 0, \dots, 6$$

$$g_m(t_i) = \sin(t_i)$$

Linear between adjacent knots.

ReLU form on $[-\pi, \pi]$ $g_m(x) =$

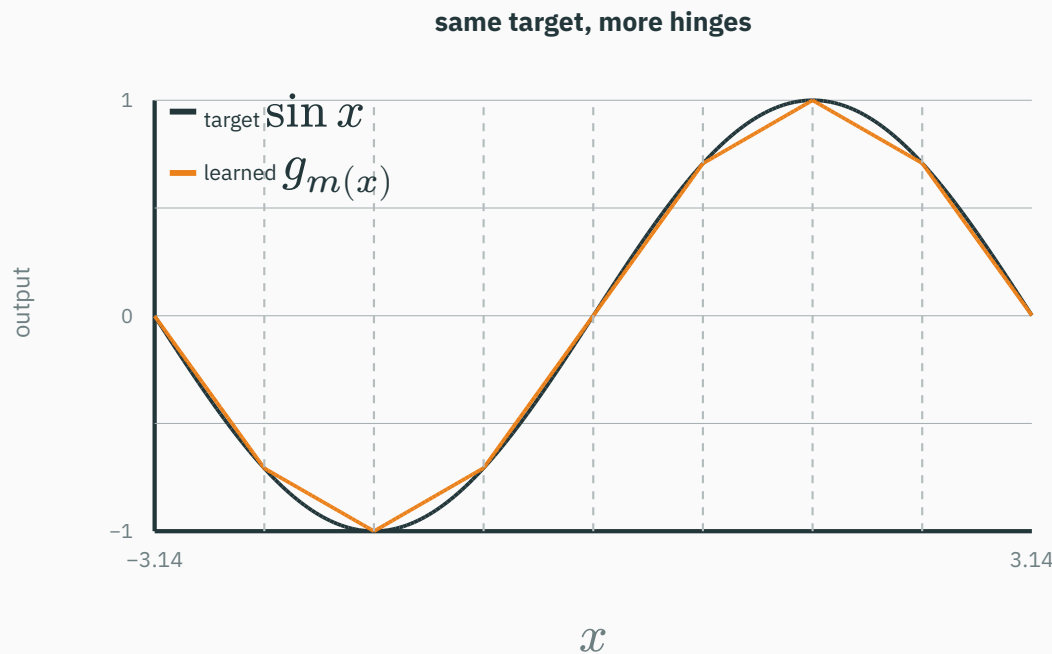
$$a + bx + \sum_{j=1}^m c_j \text{ReLU}(x - t_j)$$

Worst vertical gap $\varepsilon_m =$

$$\sup_x |\sin x - g_m(x)| \approx 0.134$$

More learned hinges follow the target for shorter distances at a time.

Seven hinges reduce the gap



LEARNED FUNCTION

Here: **7 interior breakpoints**.

$$t_i = -\pi + \frac{2\pi i}{8}, \quad i = 0, \dots, 8$$

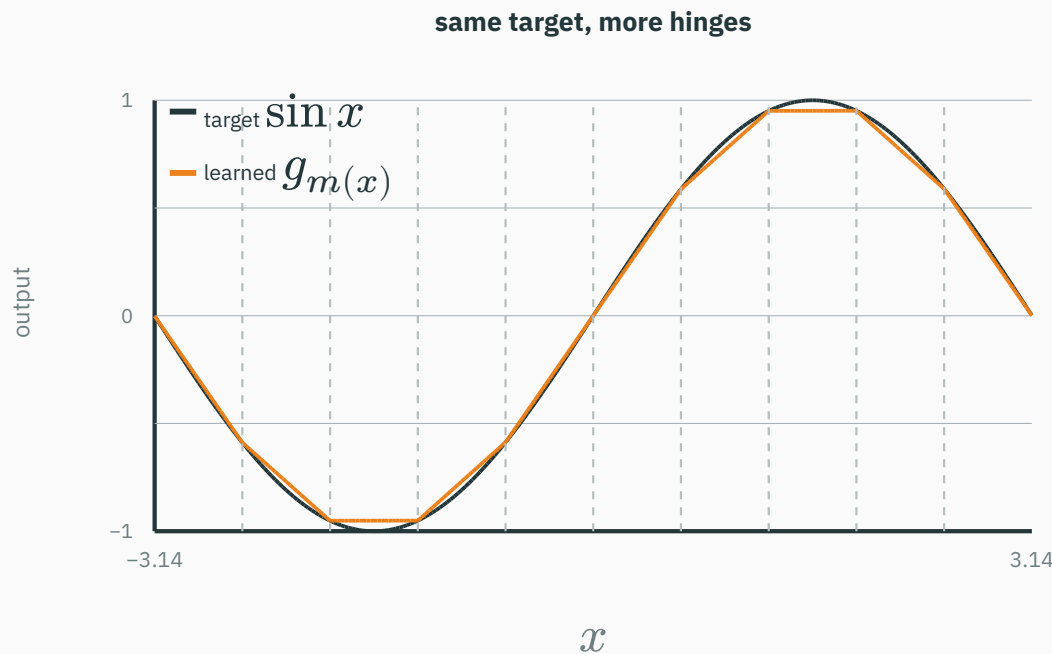
$$g_m(t_i) = \sin(t_i)$$

Linear between adjacent knots.

$$\text{ReLU form on } [-\pi, \pi] \quad g_m(x) = a + bx + \sum_{j=1}^m c_j \text{ReLU}(x - t_j)$$

$$\text{Worst vertical gap } \varepsilon_m = \sup_x |\sin x - g_m(x)| \approx 0.07$$

The worst vertical gap has fallen below 0.1.



LEARNED FUNCTION

Here: **9 interior breakpoints**.

$$t_i = -\pi + \frac{2\pi i}{10}, \quad i = 0, \dots, 10$$

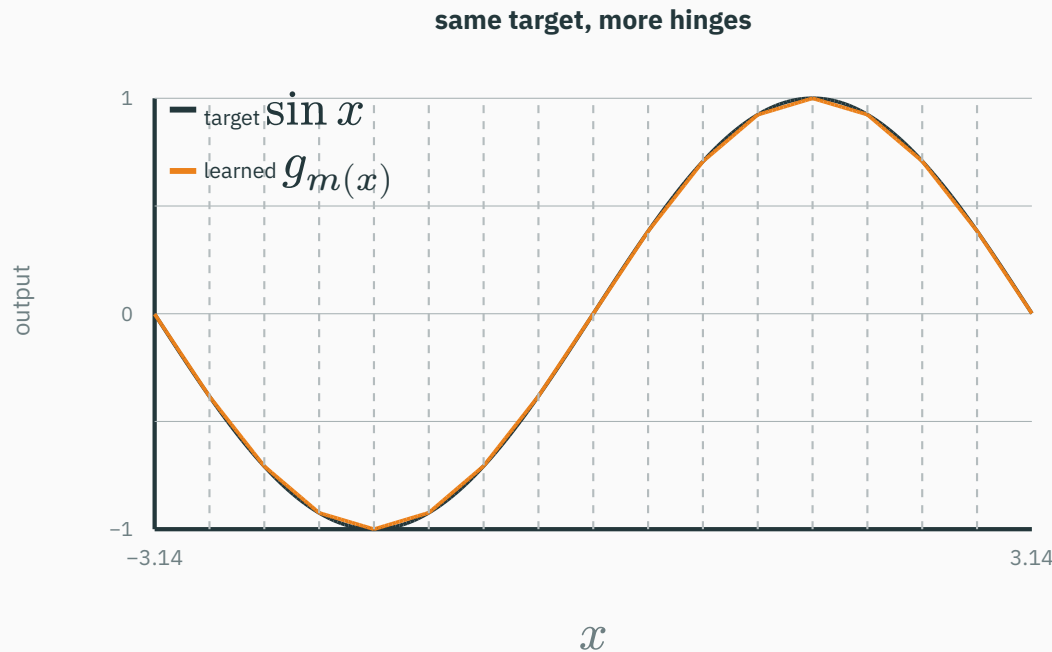
$$g_m(t_i) = \sin(t_i)$$

Linear between adjacent knots.

$$\text{ReLU form on } [-\pi, \pi] \quad g_m(x) = a + bx + \sum_{j=1}^m c_j \text{ReLU}(x - t_j)$$

$$\text{Worst vertical gap } \varepsilon_m = \sup_x |\sin x - g_m(x)| \approx 0.049$$

With nine hinges, the worst gap is already below 0.05.



LEARNED FUNCTION

Here: **15 interior breakpoints.**

$$t_i = -\pi + \frac{2\pi i}{16}, \quad i = 0, \dots, 16$$

$$g_m(t_i) = \sin(t_i)$$

Linear between adjacent knots.

$$\text{ReLU form on } [-\pi, \pi] \quad g_m(x) = a + bx + \sum_{j=1}^m c_j \text{ReLU}(x - t_j)$$

$$\text{Worst vertical gap } \varepsilon_m = \sup_x |\sin x - g_m(x)| \approx 0.019$$

Increasing width adds hinges; here the worst gap falls from 1.000 to about 0.019.

Learned ReLU units are the theorem's building blocks

For a one-dimensional input, a trained hidden unit has the form

$$h_{j(x)} = \text{ReLU}(w_j x + b_j), \quad g(x) = c + \sum_j v_j h_{j(x)}.$$

j	hinge t_j	w_j	b_j	output v_j	signed contribution $g_{j(x)}$
1	0	1	0	+1	+ReLU(x)
2	1	1	-1	-2	-2ReLU($x - 1$)
3	2	1	-2	+1	+ReLU($x - 2$)

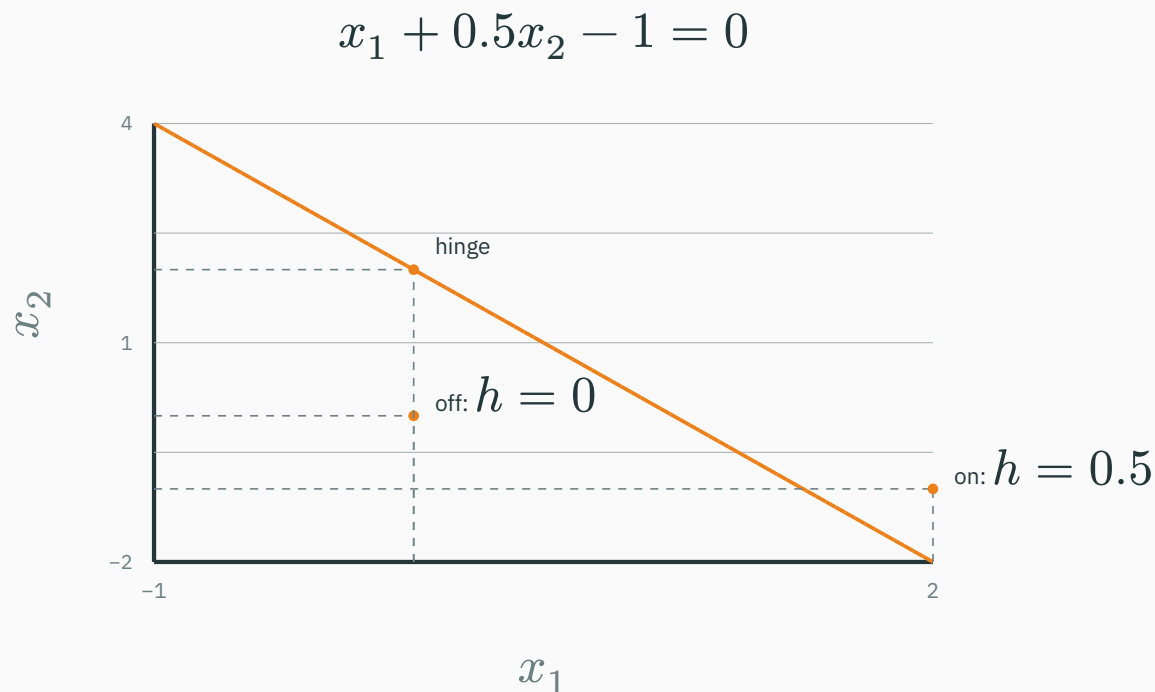
Training learns hinge locations and signed contributions; universal approximation says enough such units can make the remaining gap arbitrarily small.

In many dimensions, a hinge becomes a hyperplane

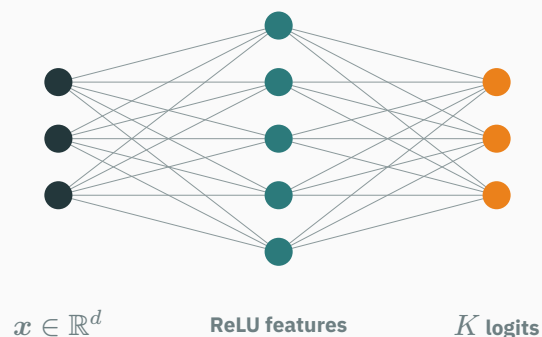
A unit reads the whole vector:

$h(\mathbf{x}) = \text{ReLU}(\mathbf{w}^\top \mathbf{x} + b)$. The hinge is where the pre-activation is zero: $\mathbf{w}^\top \mathbf{x} + b = 0$.

In 1-D this is a point. In 2-D it is a line. In 3-D it is a plane. In d dimensions it is a hyperplane.



The one-dimensional breakpoint construction generalizes to learned, oriented folds of a multidimensional input space.



1 • Vector input $\rightarrow$ learned folds

$$h_{j(x)} = \text{ReLU}(w_j^\top x + b_j).$$

2 • Features $\rightarrow$ continuous scores

$$z_{k(x)} = c_k + \sum_j v_{kj} h_{j(x)}.$$

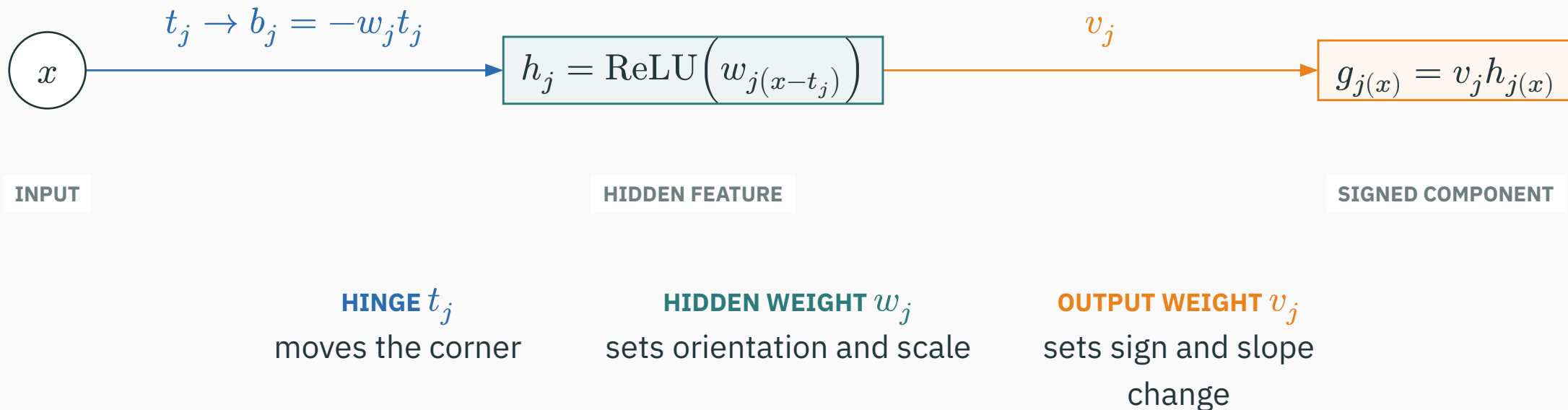
3 • Scores $\rightarrow$ classification

$$p = \text{softmax}(z), \quad \hat{y} = \arg \max_k z_k.$$

A class changes where two top scores tie.

On compact $D \subset \mathbb{R}^d$, approximate continuous logits; $\arg \max$ classifies wherever the winning score has a margin.

One hidden unit has three controls



The controls are the learned parameters; toggling components reveals how their signed sum creates the curve.

INTERACTIVE

↗ nipunbatra.github.io/dl-teaching/interactives/relu-function-builder.html

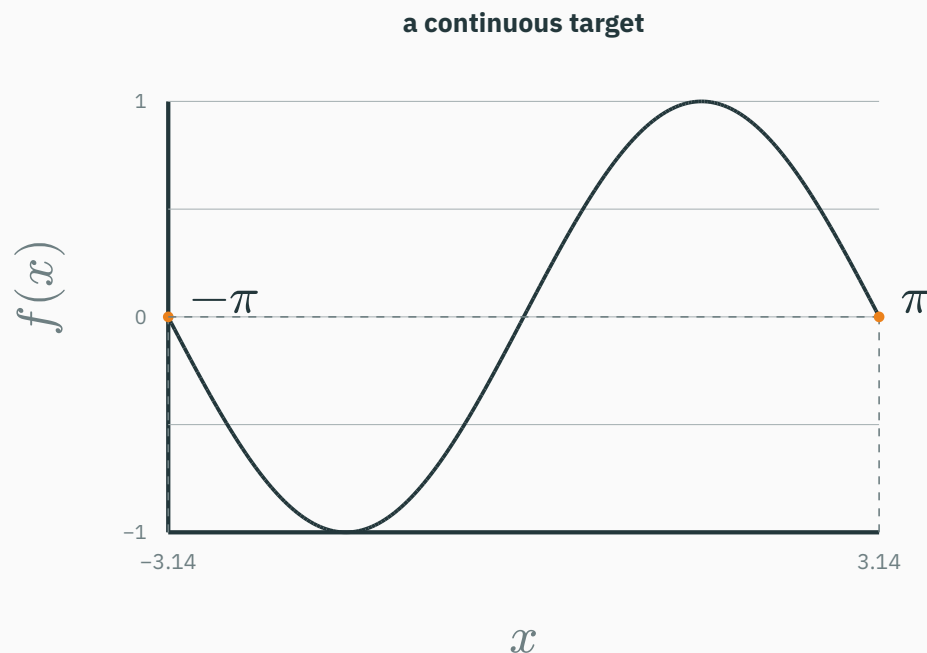
Follow the same sequence as the slides: **one hinge** → **tent** → **smooth target** → **binary classifier**.

Select one unit, then change t_j (hinge), w_j (hidden slope/orientation), or v_j (signed output contribution).

In the classifier preset, the weighted sum is the logit $z(x)$; the decision changes where $z = 0$, and $p = \sigma(z)$ reports confidence.

Inspect individual $v_j h_{j(x)}$ components first, their sum second, and the class decision last.

First fix the target and the input region



Continuous means that the target has no jumps.

In one dimension, a **compact domain** can be a closed, bounded interval such as

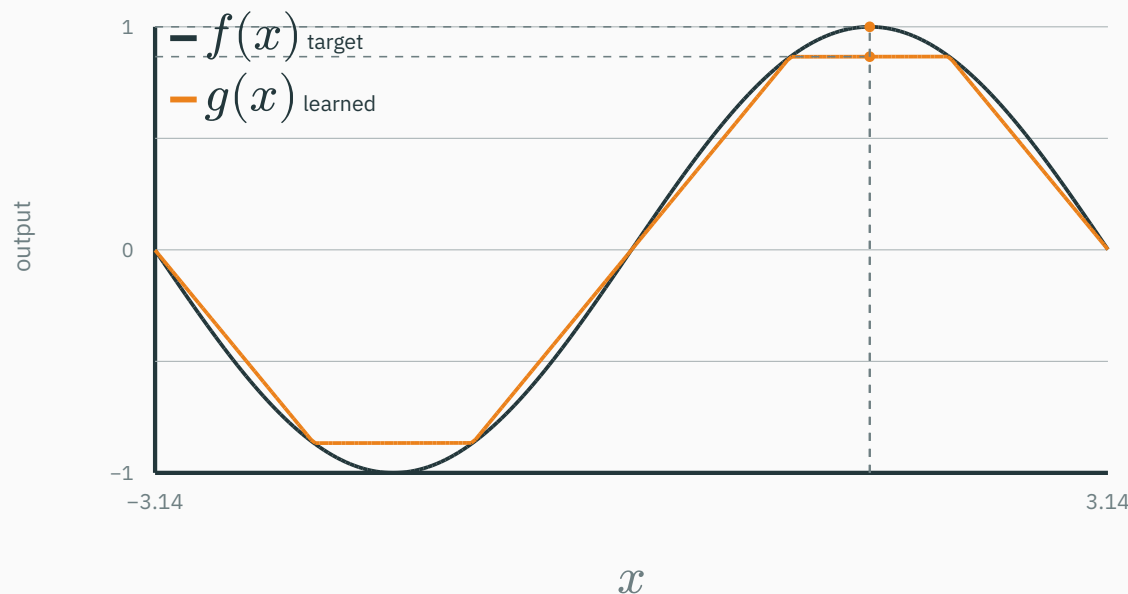
$$D = [-\pi, \pi].$$

We ask the network to match f **on** D ; the theorem makes no claim outside that chosen region.

Approximation begins with one target function on one specified input region.

$$e(x_0) = |f(x_0) - g(x_0)|$$

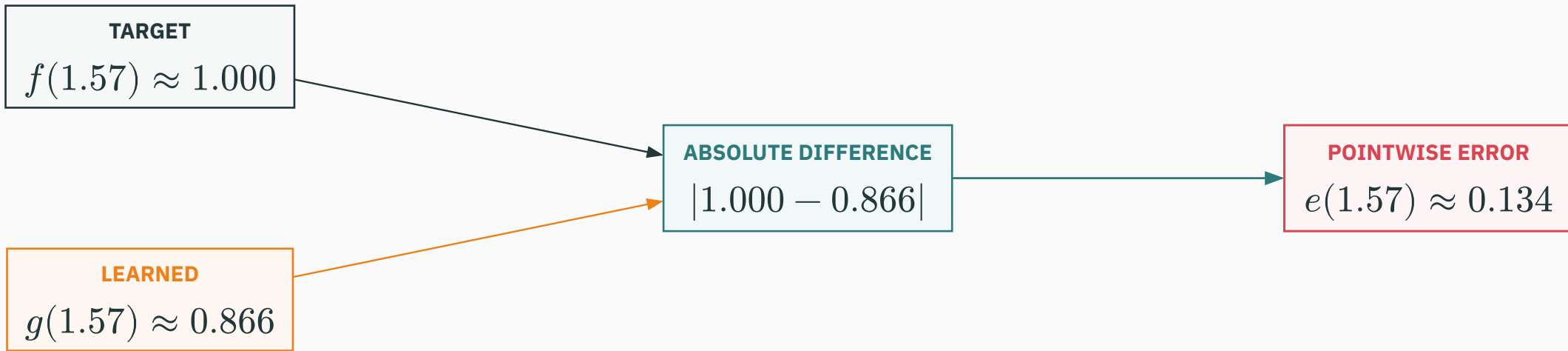
vertical slice at $x_0 = 1.57$



Pointwise error is the vertical distance between target and learned outputs at the same input.

Calculate the gap at $x_0 = 1.57$

$$e(x_0) = |f(x_0) - g(x_0)|$$



One point is not enough: universal approximation asks for the largest gap over the whole region.

“sup” means the worst gap over the whole region

$$\sup_{x \in D} |f(x) - g(x)|$$

sup means **supremum**: the smallest number at least as large as every pointwise error.

Here the error is continuous and $D = [-\pi, \pi]$ is closed and bounded, so a highest error exists. In this plot, read **sup** as **maximum**.

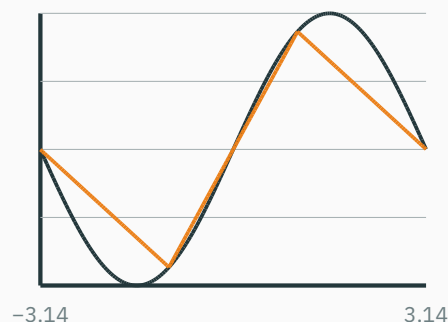
sup $< \epsilon$: no input in D crosses the tolerance.



The entire orange error curve stays below $\epsilon = 0.05$, so the uniform guarantee is satisfied.

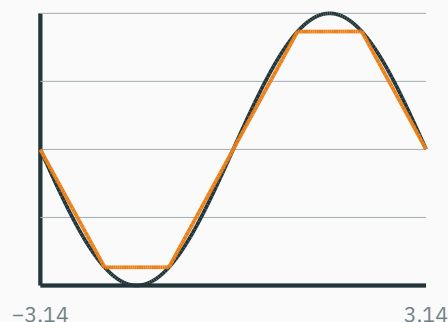
More hinges reduce the largest approximation error

4 knots



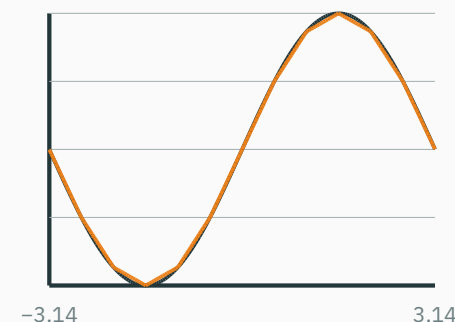
largest error ≈ 0.44

7 knots



largest error ≈ 0.13

13 knots

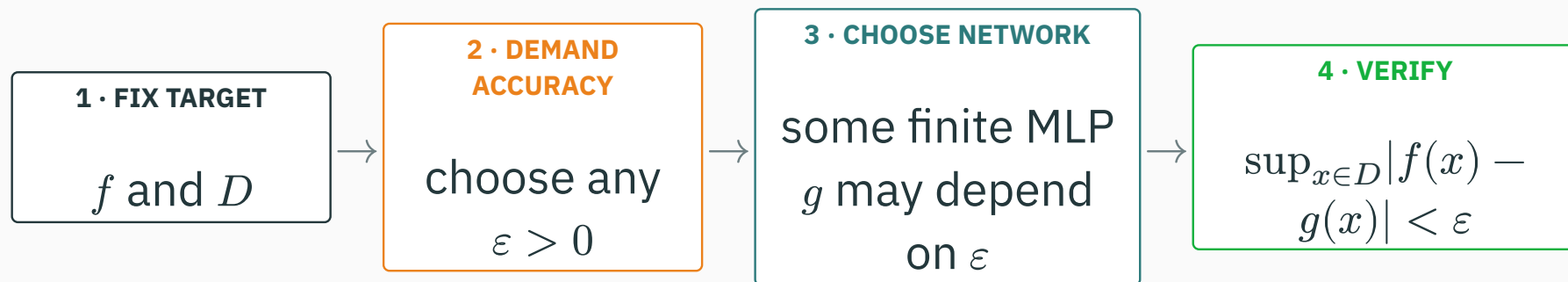


largest error ≈ 0.033

More hidden units provide more breakpoints, so a piecewise-linear network can follow finer detail.

Read the universal-approximation claim in order

D DERIVATION



$\forall \varepsilon > 0, \exists \text{ finite MLP } g \text{ such that } \sup_{x \in D} |f(x) - g(x)| < \varepsilon.$

For every ε : we may request any positive tolerance.

There exists g : the network may change when the tolerance changes.

This is a statement about representational capacity—not yet about how to find the network.

Existence does not guarantee learnability

Universal approximation does **not** promise:

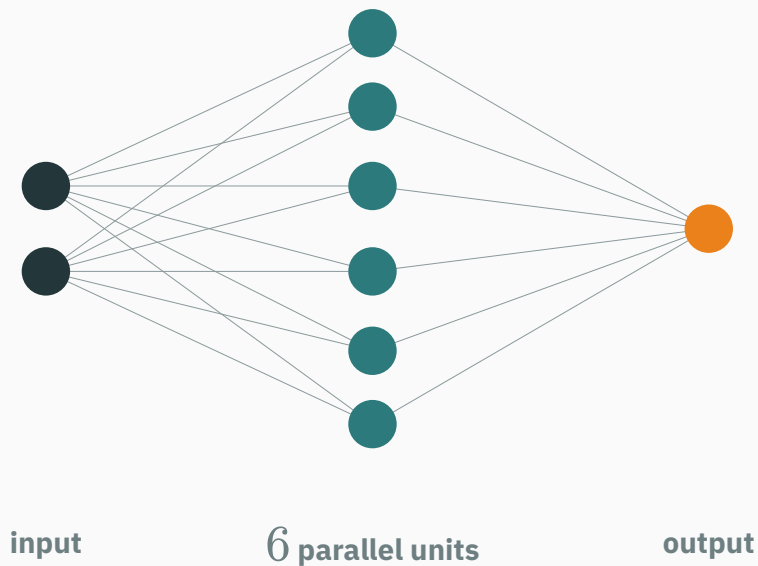
- that training will **find** that function;
- that **finite data** is enough;
- that a **small** network suffices;
- that it will **generalize**.

expressivity $\neq$ trainability $\neq$ generalization

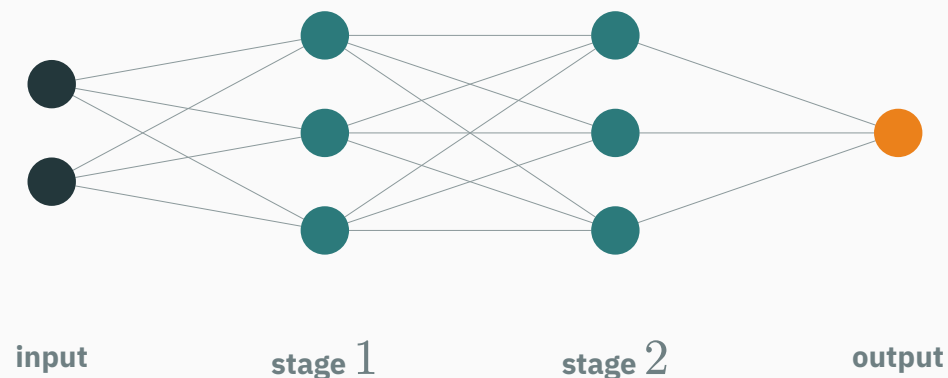
Depth versus width

Node view: width adds rows; depth adds columns

WIDER

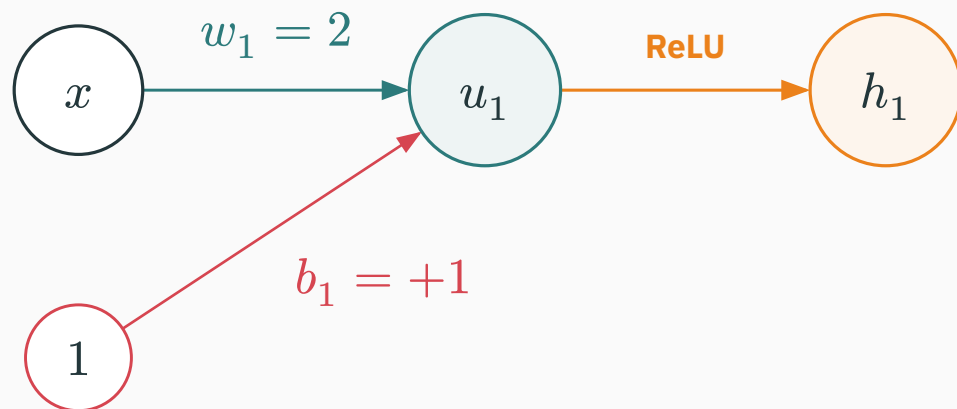


DEEPER



Count nodes vertically to read width; count hidden columns horizontally to read depth.

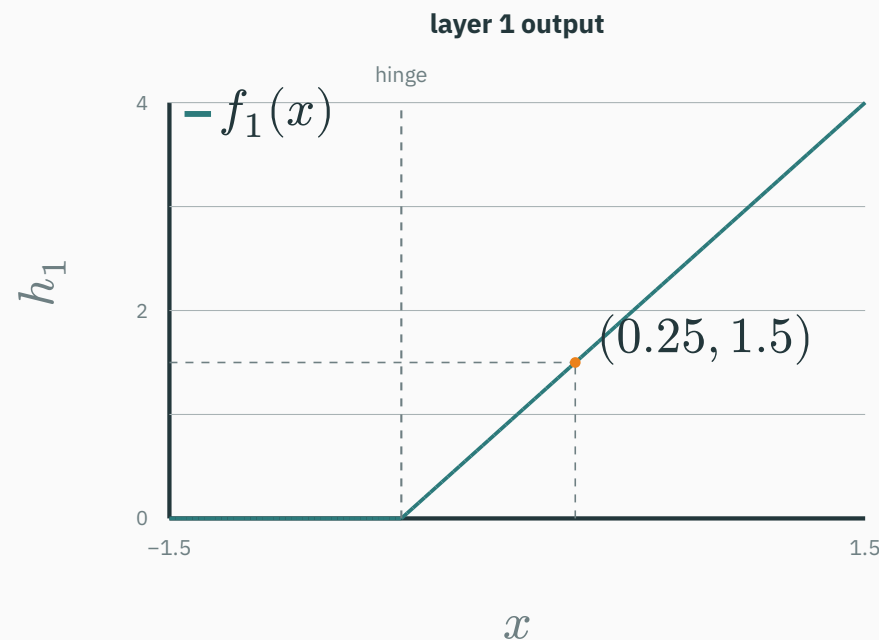
Layer 1 creates a hinge at $x = -0.5$



$$u_1 = 2x + 1$$
$$h_1 = f_1(x) = \text{ReLU}(u_1)$$

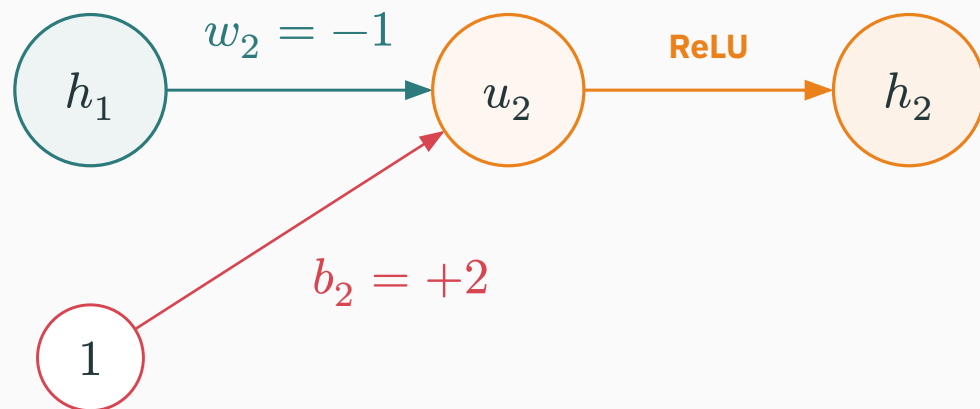
At $x = 0.25$:

$$u_1 = 1.5 \rightarrow h_1 = 1.5$$



Layer 1 is flat until $2x + 1$ becomes positive; then its learned feature grows linearly.

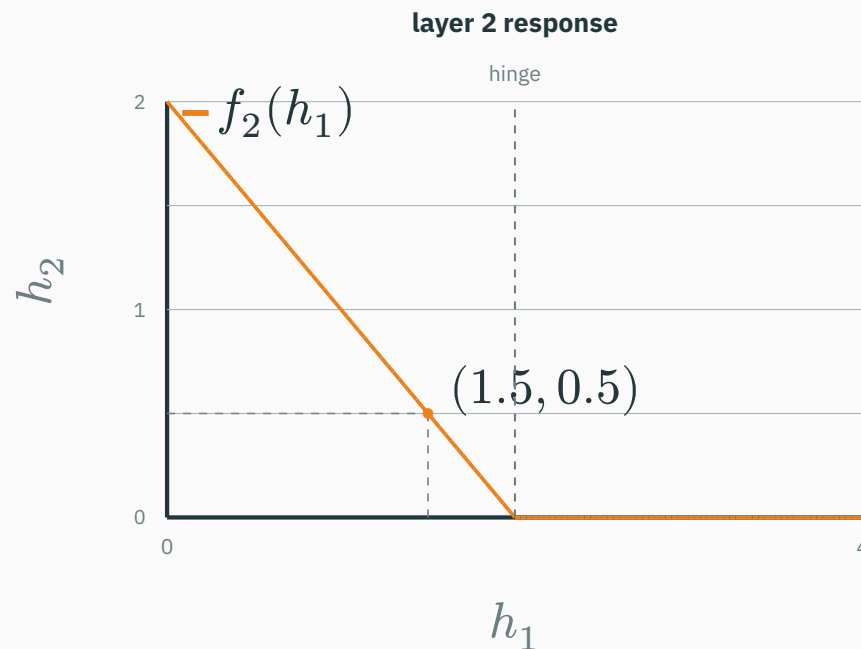
The second layer transforms the learned feature



$$u_2 = 2 - h_1$$
$$h_2 = f_2(h_1) = \text{ReLU}(u_2)$$

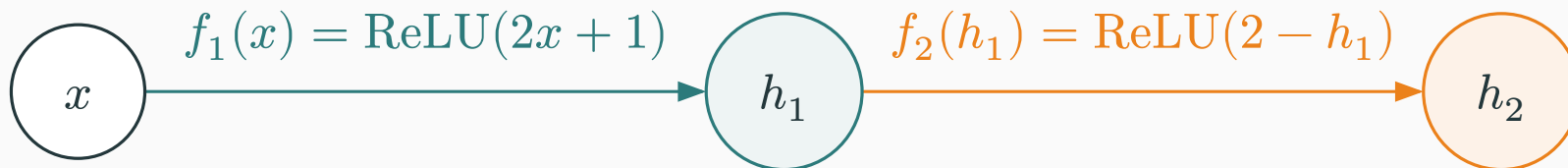
For $h_1 = 1.5$:

$$u_2 = 0.5 \rightarrow h_2 = 0.5$$

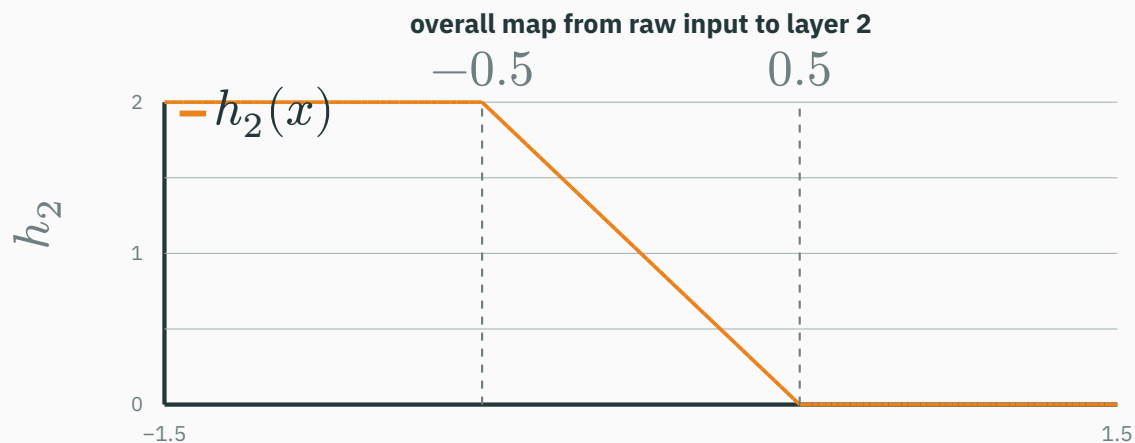


Layer 2 decreases as h_1 grows, then ReLU clips every negative result to zero.

Two layers create a clipped nonlinear ramp

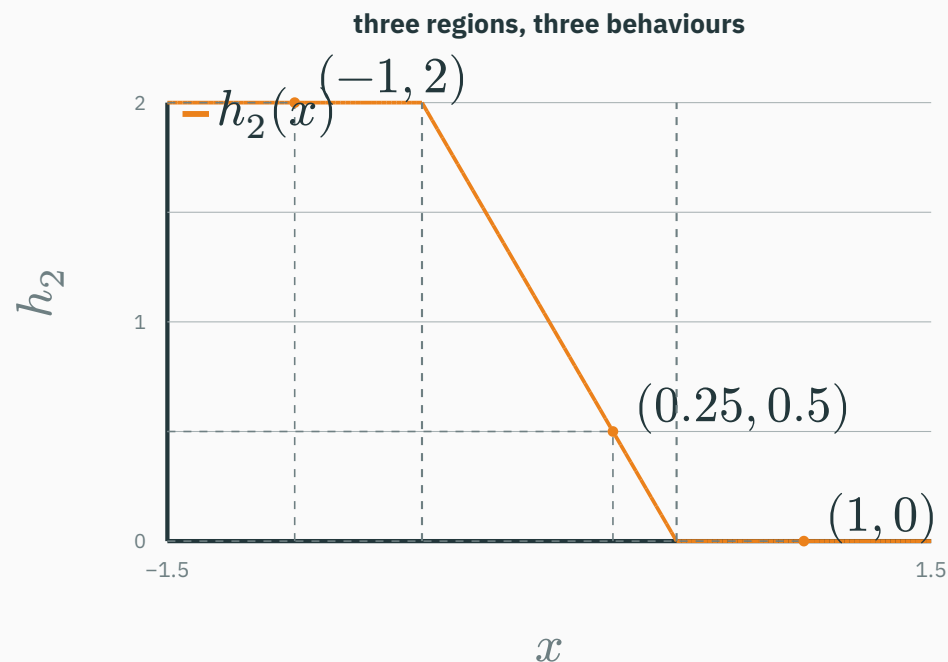


$$h_2(x) = f_2(f_1(x)) = \text{ReLU}(2 - \text{ReLU}(2x + 1))$$



The first hinge turns on at $x = -0.5$; the second layer creates another bend at $x = 0.5$.

Three inputs reveal the complete composition



$$x = -1 \quad u_1 = -1 \rightarrow h_1 = 0$$

$$u_2 = 2 \rightarrow h_2 = 2$$

$$x = 0.25 \quad u_1 = 1.5 \rightarrow h_1 = 1.5$$

$$u_2 = 0.5 \rightarrow h_2 = 0.5$$

$$x = 1 \quad u_1 = 3 \rightarrow h_1 = 3$$

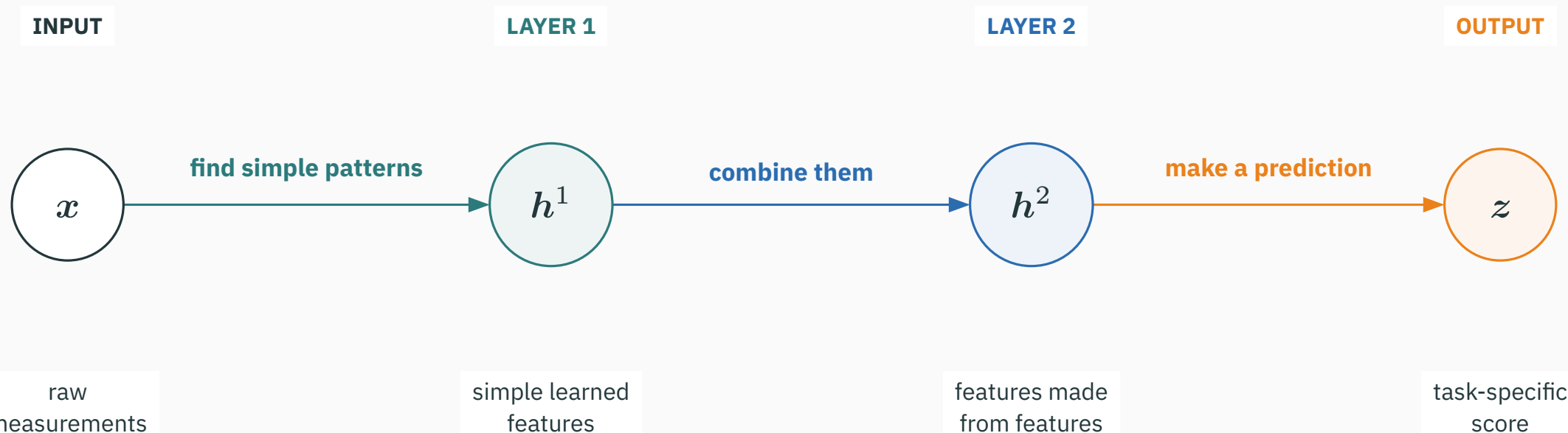
$$u_2 = -1 \rightarrow h_2 = 0$$

Depth means sequential reuse: layer 2 receives the learned value h_1 and transforms it again.

Depth builds features one stage at a time

Each layer receives the representation produced immediately before it.

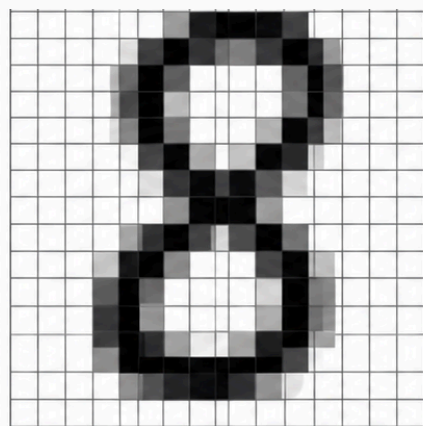
pixels → **short strokes** → **closed loops** → **digit score**



Later layers build on earlier features; they do not solve the whole task from raw input again.

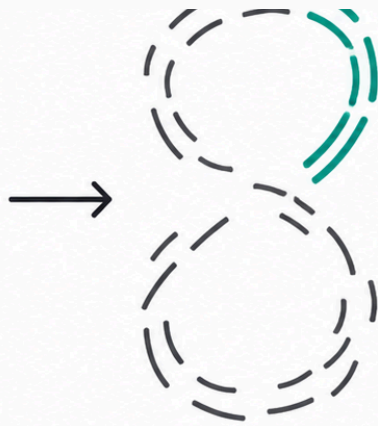
Vision: depth builds strokes, then loops

CONCEPTUAL SCHEMATIC · illustrates a possible hierarchy; not measured model activations



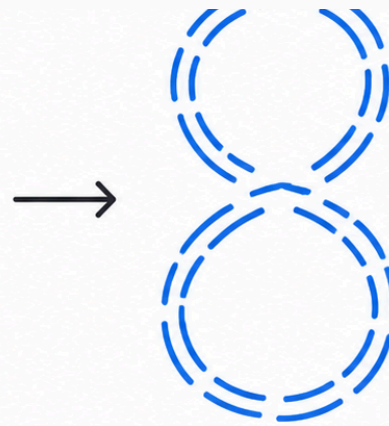
INPUT

pixel intensities



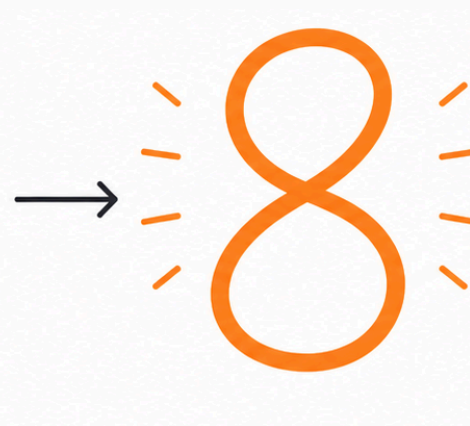
LAYER 1

curved stroke



LAYER 2

two closed loops



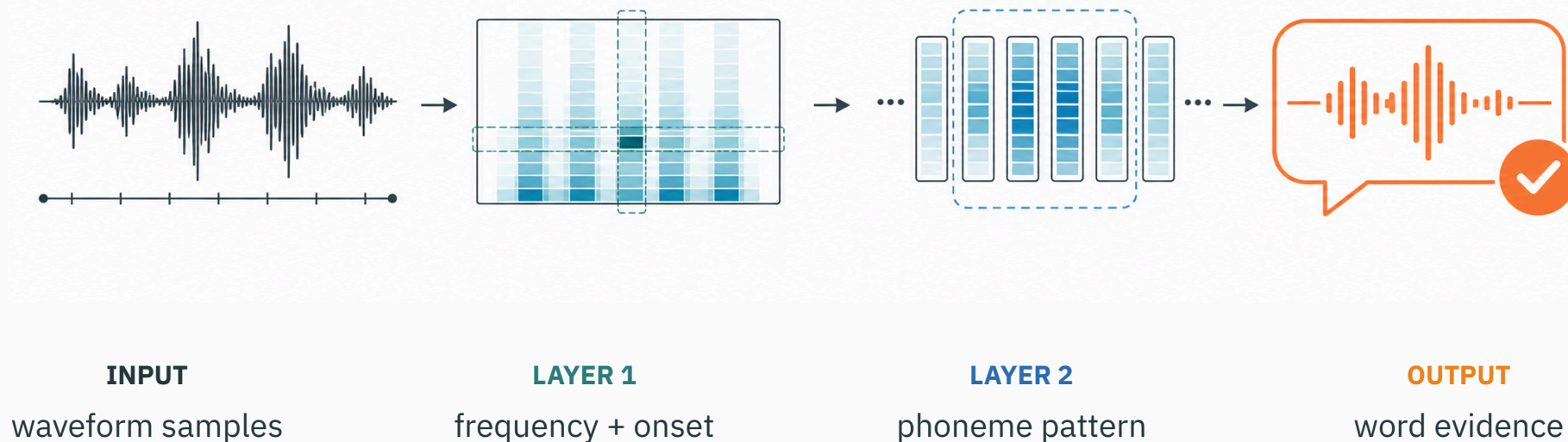
OUTPUT

evidence for 8

Early units respond to local edges; later units combine them into shapes that support the final class.

Audio: depth builds tones, then phonemes

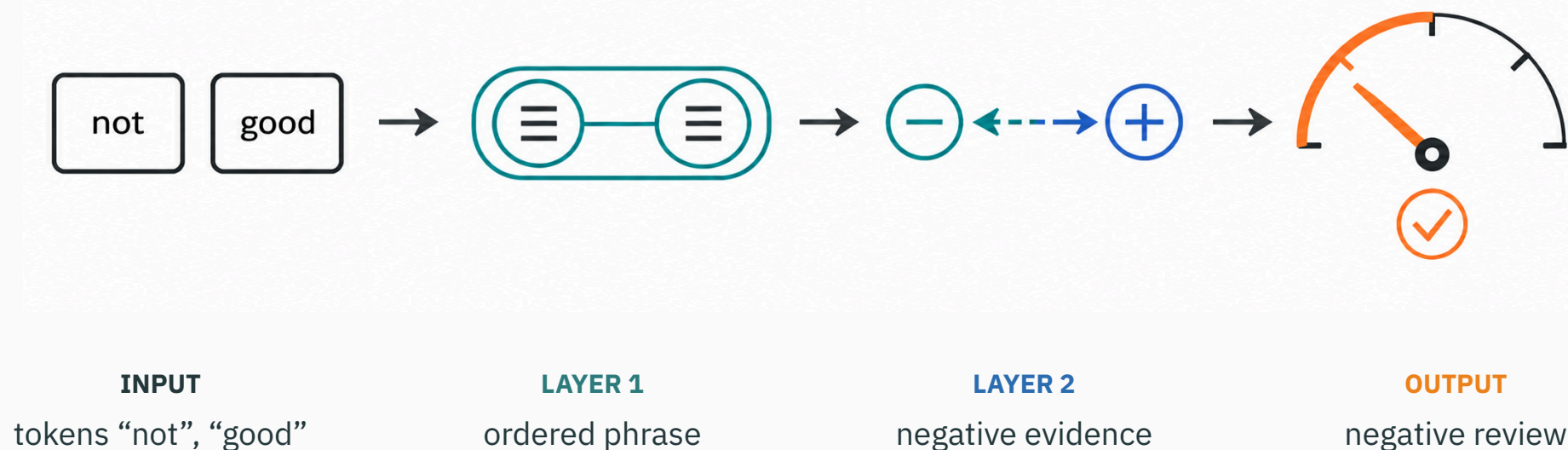
CONCEPTUAL SCHEMATIC · illustrates a possible hierarchy; not measured model activations



Early layers detect local frequency and timing cues; later layers combine them over longer time windows.

Text: depth combines words into phrase meaning

CONCEPTUAL SCHEMATIC · illustrates a possible hierarchy; not measured model activations



A later unit can reverse the evidence from "good" after combining it with the preceding token "not".

ADD WIDTH

- adds more features **in parallel**;
- helps when many patterns live at the same abstraction level;
- costs wider activations and weight matrices.

Example: add hinge locations to follow a detailed curve.

ADD DEPTH

- adds more transformations **in sequence**;
- helps when the target is compositional or hierarchical;
- costs longer paths and harder optimization.

Example: strokes → loops → digit identity.

Modern networks usually need both: enough width per stage and enough depth to compose stages.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/mlp-decision-boundary

Train a linear classifier or a small MLP on **XOR / circles / spirals / moons**.

Compare one versus two hidden layers, then adjust width while watching the learned boundary. Controls: dataset · linear/MLP · hidden width · one/two layers · activation · train/reset/resample.

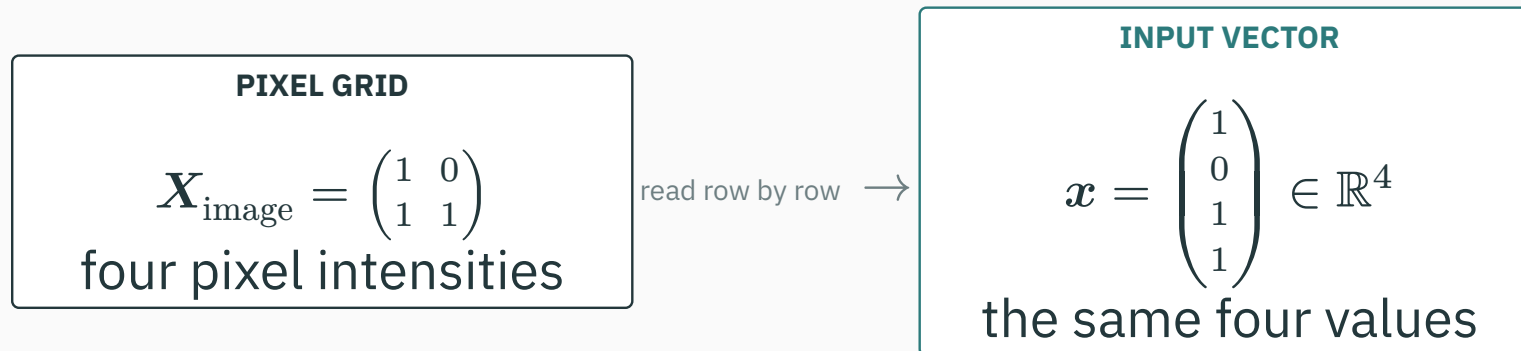
Hold the dataset fixed. First add width within one layer; then redistribute a similar number of hidden units across two layers. Which boundary appears sooner and which trains more reliably?

The comparison is empirical: width and depth change both representational structure and optimization behaviour.

Connecting to deep-learning practice

Forward pass, step 1: flatten the image into a vector

Start with a 2×2 toy image so that every number stays visible.



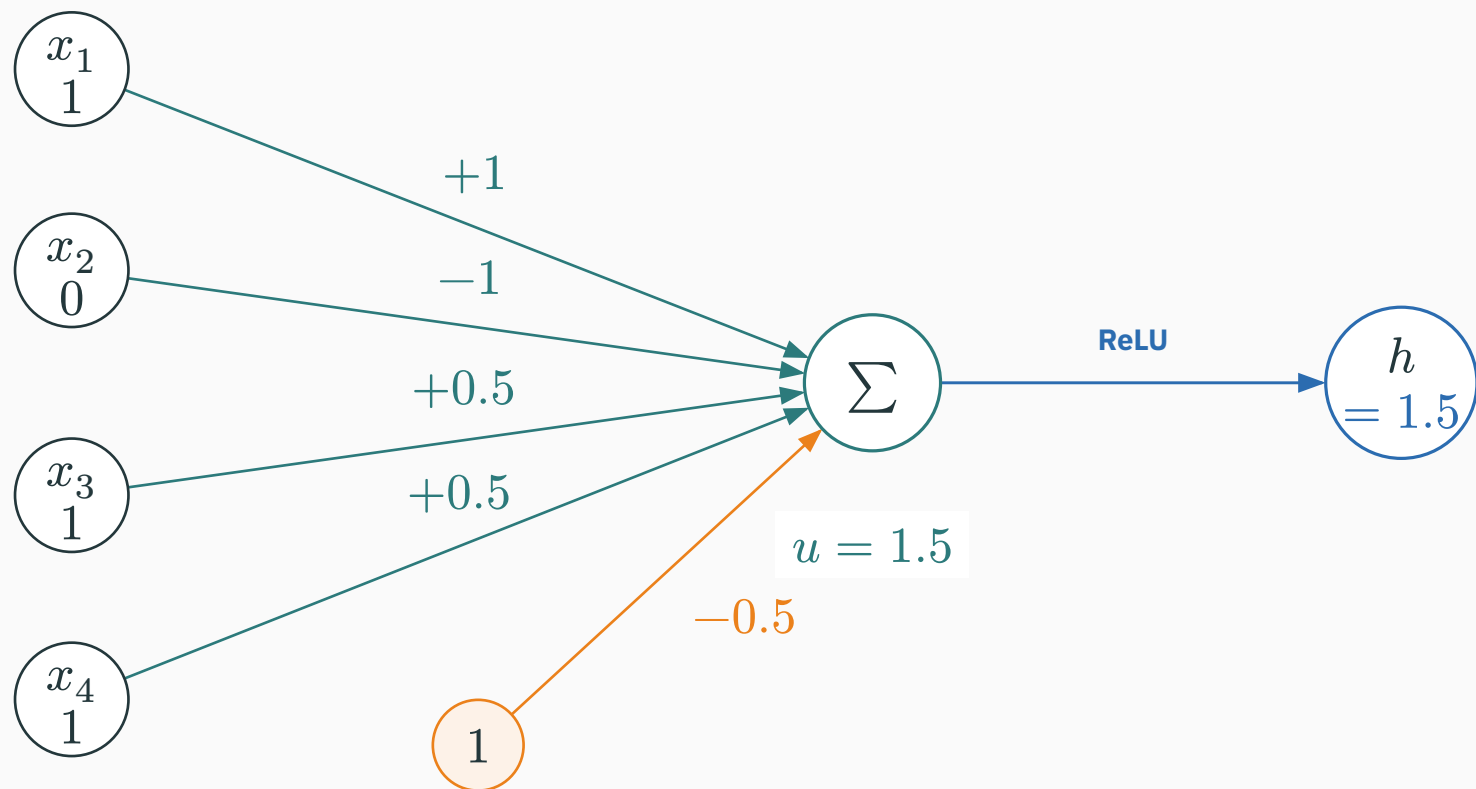
REAL IMAGE

a 28×28 image becomes $x \in \mathbb{R}^{784}$ because
 $28(28) = 784$

Flattening changes the arrangement of the pixels—not their values or meaning.

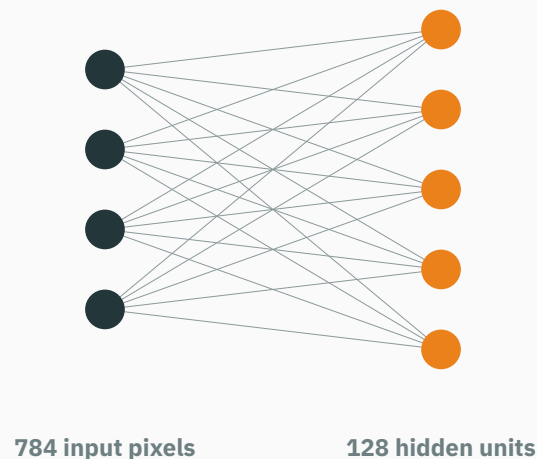
Forward pass, step 2: one hidden unit tests one pattern

D DERIVATION



$u = 1(1) - 1(0) + 0.5(1) + 0.5(1) - 0.5 = 1.5$, then $h = \max(0, u) = 1.5$. The edge weights define the pixel pattern this feature detects.

Forward pass, step 3: 128 units test 128 patterns



Each receiving unit has its own row of weights:

$$u_j = \mathbf{w}_j^\top \mathbf{x} + b_j$$
$$h_j = \max(0, u_j)$$

$$\mathbf{u} = \mathbf{W}_1 \mathbf{x} + \mathbf{b}_1 \in \mathbb{R}^{128}$$
$$\mathbf{h} = \text{ReLU}(\mathbf{u}) \in \mathbb{R}^{128}$$

schematic: only a few nodes are drawn

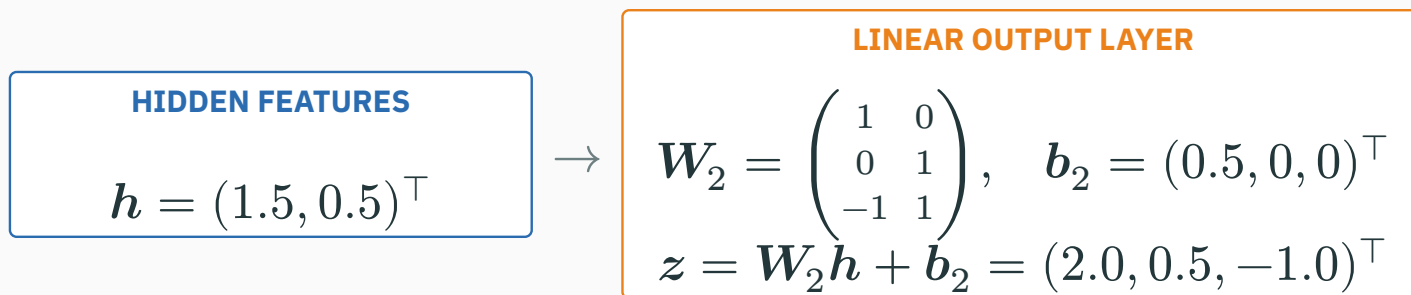
For one trained layer: $h_{17} = 2.1 \rightarrow$ curved-stroke unit **on**; $h_{83} = 0 \rightarrow$ another unit **off**.

The hidden vector is a list of 128 learned measurements of the same image.

Forward pass, step 4: hidden features become class scores

D DERIVATION

Use a three-class toy output so that every logit can be computed explicitly.



$$z_A = 1(1.5) + 0(0.5) + 0.5 = 2.0$$

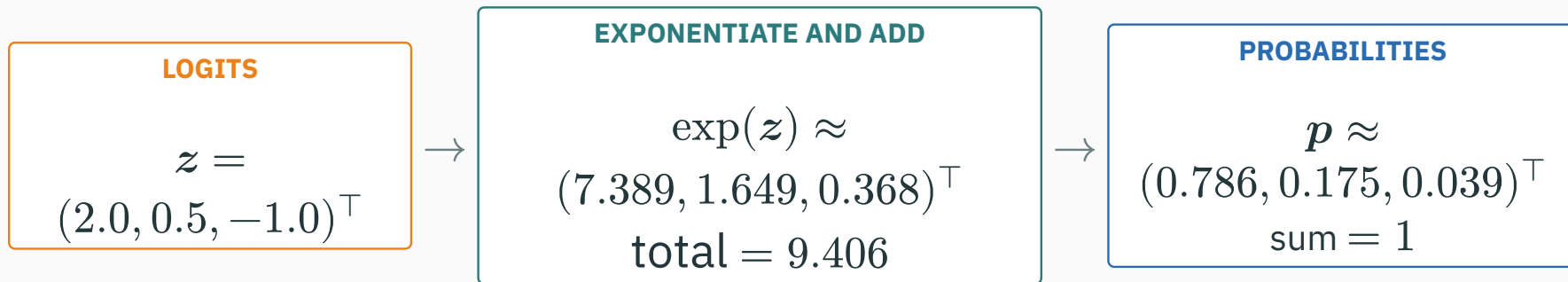
$$z_B = 0(1.5) + 1(0.5) + 0 = 0.5$$

$$z_C = -1(1.5) + 1(0.5) + 0 = -1.0$$

Logits are unrestricted class scores: larger means preferred, but they are not probabilities yet.

Forward pass, step 5: softmax compares every score

D DERIVATION



$$p_A = \frac{\exp(2.0)}{\exp(2.0) + \exp(0.5) + \exp(-1.0)} = \frac{7.389}{9.406} \approx 0.786$$

Softmax preserves the score ordering while turning all three scores into one probability distribution.

Forward pass, step 6: the label selects the loss

Cross-entropy reads the probability assigned to the true class: $\mathcal{L} = -\log p_y$.

IF $Y = A$

$$\mathcal{L} = -\log(0.786) \approx 0.241$$

IF $Y = B$

$$\mathcal{L} = -\log(0.175) \approx 1.743$$

IF $Y = C$

$$\mathcal{L} = -\log(0.039) \approx 3.244$$

High probability on the true class gives a small loss.

Confident probability on the wrong class gives a large loss.

The forward pass ends with one scalar loss; it has evaluated the weights but has not updated them.

Apply the same hidden unit from step 2 to three inputs at once.

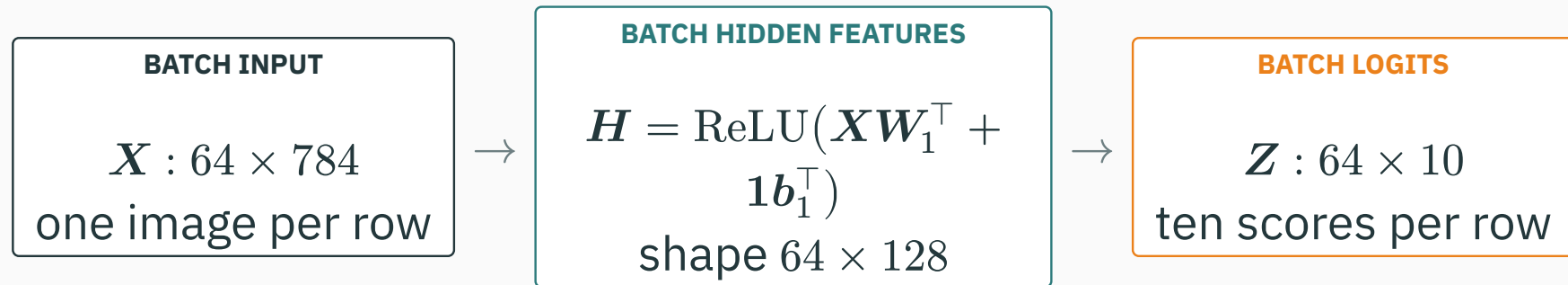
THREE INPUT ROWS $\mathbf{X} = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$ shape 3×4	$\rightarrow$	ONE FEATURE ACROSS THE BATCH $\mathbf{u} = \mathbf{X}\mathbf{w} + b\mathbf{1} = (1.5, -1, -0.5)^\top$ $\mathbf{h} = \text{ReLU}(\mathbf{u}) = (1.5, 0, 0)^\top$
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The same $\mathbf{w}$ and b are reused for row 1, row 2, and row 3.

Rows are different examples; columns are the features computed for those examples.

The full minibatch keeps the same network

For $n = 64$ images, all 128 hidden units and all 10 logits are computed together:

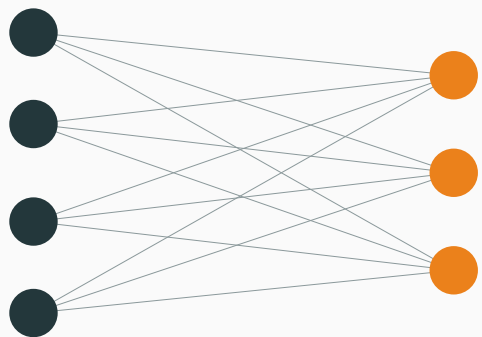


Shared across rows: weights and biases.

Recomputed for this batch: activations, probabilities, and losses.

Batching changes the array shapes, not the prediction rule: every row follows the same network.

A tiny layer reveals the parameter-count rule



4 incoming values

3 receiving units

every connecting edge carries one weight

Weights

$$4 \text{ inputs} \times 3 \text{ units} = 12$$

Biases

$$\text{one per receiving unit} = 3$$

Total

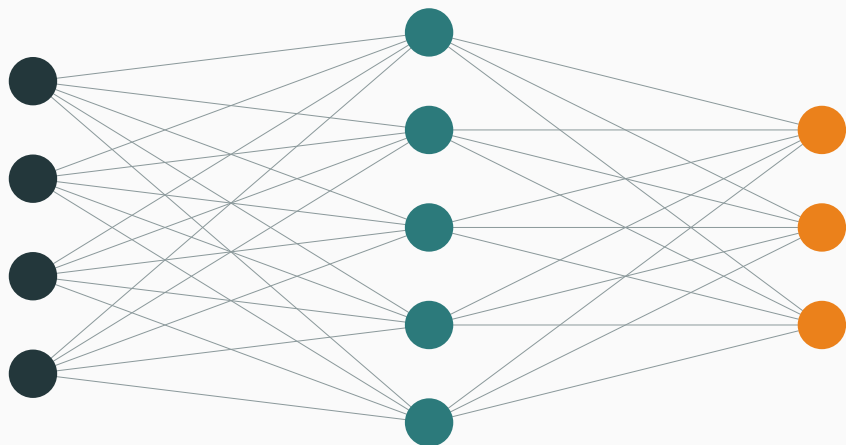
$$12 + 3 = 15 \text{ parameters}$$

GENERAL RULE

$$(\text{output width} \times \text{input width}) + \text{output width}$$

Count the arrows, then add one bias for every receiving node.

Apply the same counting rule to both affine layers



784 inputs

128 hidden units

10 outputs

The drawing is schematic; the labels give the actual widths.

INPUT → HIDDEN

$$W_1 : 128 \times 784 \rightarrow 128(784) = 100,352$$

$$b_1 \in \mathbb{R}^{128} \rightarrow 128$$

Subtotal: $100,352 + 128 = 100,480$

HIDDEN → OUTPUT

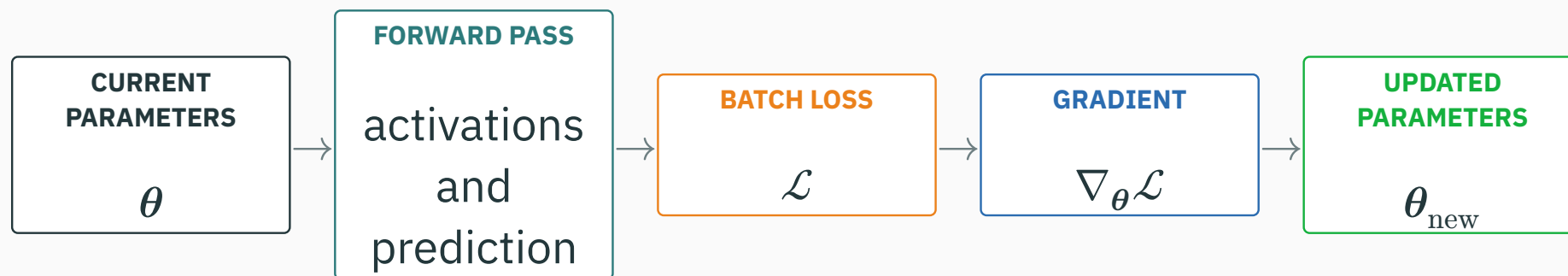
$$W_2 : 10 \times 128 \rightarrow 10(128) = 1,280$$

$$b_2 \in \mathbb{R}^{10} \rightarrow 10$$

Subtotal: $1,280 + 10 = 1,290$

Total trainable parameters: $100,480 + 1,290 = 101,770$.

Training changes parameters, not the architecture



$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla_{\theta} \mathcal{L}$$

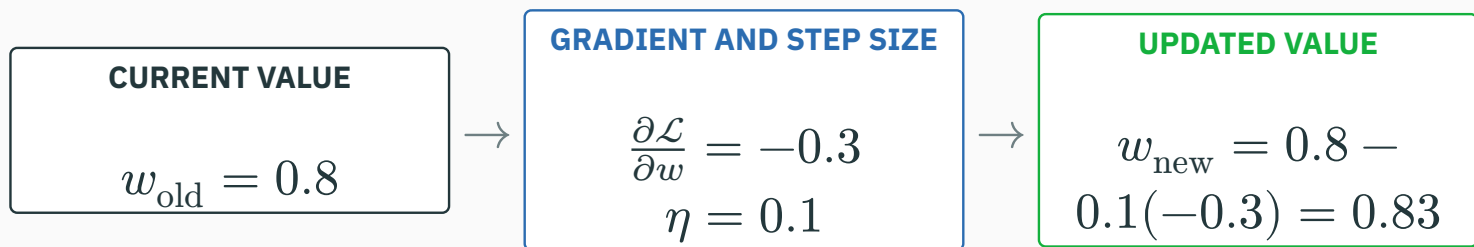
Fixed: layer sizes, connectivity pattern, and activation choices.

Changed: the entries of weight matrices and bias vectors.

Forward propagation produces activations and loss; learning uses that loss to update the shared parameters.

One numerical parameter update

D DERIVATION

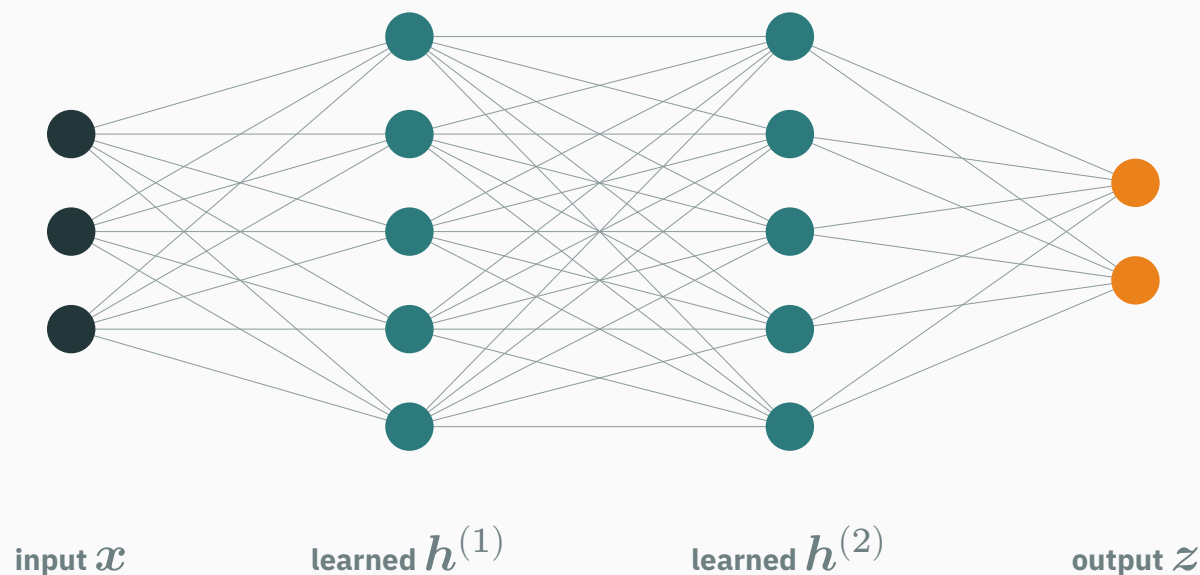


The negative gradient says increasing w locally decreases the loss, so the update moves w upward.

Lecture 3 will explain how backpropagation computes every component of $\nabla_{\theta} \mathcal{L}$ efficiently.

The forward pass evaluates the current network; the update changes the network for the next pass.

The decisive change is the learned representation



The final scores still feed the familiar Lecture 1 output model and loss.

The output loss can stay the same while hidden layers make the representation nonlinear.

We now have $x \rightarrow f_{\theta}(x) \rightarrow \mathcal{L}$.

Lecture 3: how do we compute $\nabla_{\theta}\mathcal{L}$ efficiently?

computation graphs · the chain rule · backpropagation · automatic differentiation